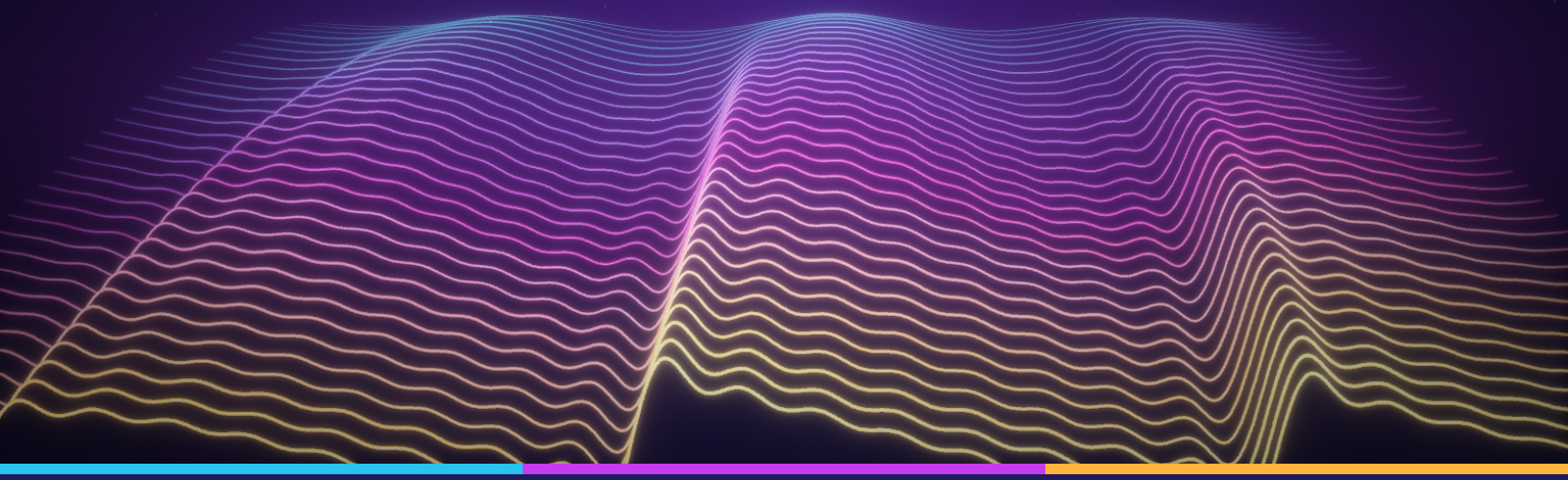
MASSIVE PSXL

Phil S Audio



Massive PSXL

User Manual and Tutorials

A wavetable synthesizer by Phil S Audio

Reference manual | 28 step-by-step tutorials | 117 factory presets

Version 1.0.0 | Phil S Audio | VST3, Audio Unit and Standalone for macOS (Apple Silicon)

Contents

About this manual . . . 6

Part I: Reference Manual . . . 7

1 Welcome . . . 8

Feature summary . . . 8
Formats and requirements . . . 8
How a sound is made (the short version) . . . 9

2 Installation . . . 9

Installer package . . . 9
Per-user install script (no password) . . . 9
Building from source (developers) . . . 9
Gatekeeper and ad-hoc signing . . . 10
First launch . . . 10
Where Massive PSXL keeps its files . . . 10
Uninstalling . . . 10

3 Quick Start . . . 11

Step by step . . . 11
Shortcut: quick setups . . . 11
Make it move . . . 12
Or just load a preset . . . 12
No sound? Checklist . . . 12

4 Interface Tour . . . 12

4.1 The top bar . . . 14
4.2 The Menu . . . 14
4.3 Knobs: the basics . . . 14
4.4 Pitch panel . . . 15
4.5 Oscillator panel . . . 15
4.6 Noise, Filter, Insert, Amp and FX panels . . . 15
4.7 The lower tab strip . . . 15
4.8 The macro strip and keyboard . . . 16
4.9 Toasts and feedback . . . 16

5 Signal Flow and Routing . . . 16

5.1 The seven nodes . . . 16
5.2 The one rule: left to right . . . 16
5.3 Making and removing connections . . . 17
5.4 Quick setups . . . 17

5.5 Things to know about routing . . . 17

6 Oscillators and Wavetables . . . 18

6.1 Oscillator controls . . . 18
6.2 The ten built-in wavetables . . . 18
6.3 Unison . . . 19
6.4 Phase Mod . . . 19
6.5 Fold . . . 19
6.6 Wavetable library and importing your own . . . 19

7 Noise . . . 20

Noise types . . . 20

8 Filters . . . 21

8.1 Ladder . . . 21
8.2 State Variable . . . 21
8.3 Comb . . . 21
8.4 Formant . . . 22
8.5 Using two filters . . . 22

9 Insert and Amp . . . 23

9.1 The Insert . . . 23
9.2 The Amp . . . 23

10 Master Effects . . . 24

10.1 EQ . . . 24
10.2 Distortion . . . 24
10.3 Chorus . . . 24
10.4 Delay . . . 24
10.5 Reverb . . . 25
10.6 Master Volume and safety . . . 25

11 Modulation . . . 25

11.1 How modulation adds up . . . 25
11.2 Sources . . . 26
11.3 Envelopes . . . 26
11.4 LFOs . . . 27
11.5 Performers . . . 28
11.6 Macros and performance sources . . . 29
11.7 Assigning, editing and removing . . . 29
11.8 Modulation destinations (42) . . . 29

12 Smart Macro Naming . . . 30

12.1 The rules . . . 30

12.2 Renaming by hand . . . 30
12.3 Hardware and host displays . . . 30
12.4 Macros and MIDI . . . 30

13 Voice, Glide and Tuning . . . 31
13.1 Voice modes . . . 31
13.2 Glide . . . 31
13.3 Pitch, bend and sustain . . . 31
13.4 Microtuning with Scala files . . . 31

14 Presets and the Browser . . . 32
14.1 Browsing . . . 32
14.2 The factory library . . . 32
14.3 Saving your own presets . . . 32
14.4 Init and Mutate . . . 33
14.5 Presets, programs and your DAW . . . 33

15 MIDI, Keyboard and MIDI Learn . . . 33
15.1 MIDI messages understood . . . 33
15.2 The on-screen keyboard . . . 33
15.3 MIDI Learn . . . 34

16 Themes, Scaling and Settings . . . 34
16.1 Light and dark themes . . . 34
16.2 Interface size . . . 34
16.3 Where settings are stored . . . 34
16.4 Undo and Redo . . . 35
16.5 Tooltips and messages . . . 35

17 NKS Status . . . 35
17.1 What is implemented in the plug-in . . . 35
17.2 What is not possible without the standard's owner . . . 35
17.3 What to expect today . . . 35

18 Parameter Reference . . . 36
18.1 Quick counts . . . 43

19 Mouse and Keyboard Reference . . . 43

20 Troubleshooting . . . 44

21 Glossary . . . 45

Part II: Tutorials . . . 47

How to Use the Tutorials 48

Tutorial 1: Kick drum 49
Tutorial 2: Snare drum 50
Tutorial 3: Closed and open hi-hats 51
Tutorial 4: Hand clap 52
Tutorial 5: Tom 53
Tutorial 6: Sub bass 54
Tutorial 7: Wobble bass 55
Tutorial 8: Acid bass 56
Tutorial 9: Reese bass 57
Tutorial 10: Supersaw lead 58
Tutorial 11: Mono lead (expressive) 59
Tutorial 12: Pluck 60
Tutorial 13: Bell 61
Tutorial 14: Electric piano 62
Tutorial 15: Drawbar organ 63
Tutorial 16: String pad 64
Tutorial 17: Choir and vowel pad 65
Tutorial 18: Brass stab 66
Tutorial 19: Riser and sweep 67
Tutorial 20: Laser zap 68
Tutorial 21: Wind and ocean 69
Tutorial 22: Arpeggio with a Performer 70
Tutorial 23: Trance gate with a Performer 71
Tutorial 24: Talking filter (formant) 72
Tutorial 25: FM bell with Phase Mod 73
Tutorial 26: Tempo-synced stutter 74
Tutorial 27: Build your own macro-controlled patch 75
Tutorial 28: Import your own wavetable 76

A Factory Presets 77

About this manual

This manual documents Massive PSXL version 1.0.0 by Phil S Audio exactly as the plug-in is built. Part I is the reference manual. Part II is a set of step-by-step tutorials that start from **Load Init (Blank)** and build real sounds. Appendix A lists the 117 factory presets.

Conventions

- **Bold** text is something you can see on screen: a button, tab, knob label, menu item or panel.
- **Menu > Settings > Interface size** means: press the Menu button, open the Settings sub-menu, then choose Interface size.
- Knob values are written the way the knob shows them: 0 to 1 amounts, Hz, seconds, semitones (st) or cents (ct).
- Modulation amounts run from -1 to +1 and are fractions of the target knob's full travel (chapter 11).
- Middle C is MIDI note 60 and is labelled C3 on the on-screen keyboard.
- Values in tutorials are **starting points**: adjust them by ear.

> **NOTE.** The interface drawings in this manual are schematic illustrations drawn from the plug-in's layout, not screenshots. Names, tab order and menu wording follow the plug-in's source code.

What is in the box

- **Massive PSXL.vst3** (VST3 plug-in), **Massive PSXL.component** (Audio Unit) and **Massive PSXL.app** (standalone), built for Apple Silicon Macs.
- 117 original factory presets in 8 categories (Bass, Leads, Pads, Plucks, Keys, Drums, FX, Sequence).
- 10 built-in wavetables, plus 22 slots for your own.

PART I

Reference Manual

Installation, the interface, sound generation, modulation, presets, MIDI and a complete parameter reference.

1 Welcome

What Massive PSXL is and what it can do

Massive PSXL is a wavetable synthesizer. Two oscillators scan through tables of waveforms, three sources (Oscillator 1, Oscillator 2 and a noise generator) feed a patchable chain of two filters and an insert effect, and a master effects section finishes the sound. Everything you hear can be shaped by envelopes, LFOs, step-sequence Performers, macros and performance controllers.

The design idea is a **visible signal path**. Instead of a fixed chain, the **Routing** tab is a node editor: you see the oscillators, filters and insert as boxes and connect them with wires that glow with the audio passing through them.

Feature summary

Area	What you get
Oscillators	2 wavetable oscillators. Unison of 1 to 8 voices with detune, Phase Mod (each oscillator can phase-modulate the other), Fold, Pitch +/-48 st, Fine +/-100 ct.
Wavetables	10 built-in tables of 32 frames each; import WAV, AIFF, FLAC or OGG files into 22 user slots. Band-limited playback.
Noise	White, Pink, Brown and Crackle noise with a low-pass/high-pass Color control.
Filters	2 filters, each with 4 models (Ladder, State Variable, Comb, Formant) and 4 modes per model, plus Drive and Key Track.
Insert	6 per-voice shapers: TanH, Fold, Bitcrush, Hard Clip, Rectify, Sine.
Routing	Node editor: any source can feed any later node (Osc 1/2/Noise to Filter 1, Filter 2, Insert, Out). Per-wire gain up to 150 percent.
Amp	Amp envelope, Level, Pan, Feedback and velocity sensitivity.
Master FX	EQ (low shelf, mid peak, high shelf), Distortion, Chorus, tempo-syncable stereo Delay, Reverb.
Modulation	Amp envelope + 8 slots (each an envelope or LFO), 3 Performers (16-step sequencers), 16 Macros, Velocity, Key Track, Random, Mod Wheel, Aftertouch, Pitch Bend. 42 destinations, up to 256 assignments.
Voices	Poly (1 to 16 voices), Mono and Legato modes, glide (linear or exponential), pitch bend range 0 to 24 st, sustain pedal.
Tuning	Scala (.scl) microtuning, reset to 12-tone equal temperament.
Presets	117 factory presets, searchable browser, user presets in your Music folder, Mutate (subtle / wild).
Workflow	Undo / Redo, MIDI Learn on any parameter, drag-to-assign modulation, smart macro naming, light and dark themes, resizable interface (75 to 200 percent), on-screen keyboard with pitch and mod wheels.

Formats and requirements

Item	Detail
Formats	VST3, Audio Unit (AU, instrument) and Standalone application.
Platform	macOS 11.0 or later on Apple Silicon (M1, M2, M3 and later). The default build is native arm64; the project can also be built as a universal (arm64 + x86_64) binary.
Audio	Stereo output, no audio input. Works at any sample rate your host provides.
MIDI	Receives notes, velocity, pitch bend, mod wheel (CC 1), sustain (CC 64), channel pressure and learnable CCs.
Plug-in IDs	Manufacturer code PhSA, plug-in code Psxl (used for Audio Unit validation).

How a sound is made (the short version)

1. **Sources** make raw sound: Osc 1, Osc 2 and Noise.
2. The **Routing** view sends sources through **Filter 1**, **Filter 2** and the **Insert** and finally to **Out**.
3. The **amp envelope**, level and pan turn that into a note you play. This part runs once per voice.
4. All voices are summed and pass through the **master effects** (EQ, Distortion, Chorus, Delay, Reverb) and the master Volume.
5. **Modulators** (envelopes, LFOs, Performers, macros, controllers) move the knobs while you play.

2 Installation

macOS package, per-user script and first launch

Massive PSXL is built for macOS on Apple Silicon. There are two ways to install it: the installer package (recommended) and a per-user script that needs no administrator password.

Installer package

1. Open **Massive-PSXL-1.0.0.pkg** and follow the installer. It needs an administrator password because it installs to the system-wide folders.
2. The package installs three items (below). A plain zip of the same three bundles is also produced for manual installs.
3. Restart your DAW, or rescan plug-ins, so it finds the new instrument.

Item	Installed to
VST3 plug-in	/Library/Audio/Plug-Ins/VST3/Phil S Audio/Massive PSXL.vst3
Audio Unit	/Library/Audio/Plug-Ins/Components/Massive PSXL.component (grouped under the manufacturer name Phil S Audio)
Standalone app	/Applications/Phil S Audio/Massive PSXL.app

NOTE. The VST3 goes in a **Phil S Audio** sub-folder, so in hosts that show folders it appears under that name.

Per-user install script (no password)

If you build the plug-in from source, or you just want a copy for your own user account, run the script from the project folder:

```
installer/install_local.sh [build-dir]
```

The build folder defaults to **build**. The script copies the three bundles to your home folder, signs each one with an ad-hoc signature, and restarts the Audio Unit registrar so hosts see the new plug-in:

Item	Installed to
VST3	~/Library/Audio/Plug-Ins/VST3/Phil S Audio/Massive PSXL.vst3
Audio Unit	~/Library/Audio/Plug-Ins/Components/Massive PSXL.component
Standalone	~/Applications/Phil S Audio/Massive PSXL.app

Building from source (developers)

The project uses CMake and fetches JUCE 8.0.8 automatically. On an Apple Silicon Mac:

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release --parallel
```

```
bash installer/build_installer.sh build
```

The last command produces **dist/Massive-PSXL-1.0.0.pkg** and a zip of the plug-in bundles. Add **-DPSXL_UNIVERSAL=ON** to the first command for a universal binary.

Gatekeeper and ad-hoc signing

Apple Silicon Macs refuse to load unsigned native code, so the installer script signs the bundles with an **ad-hoc signature** (identity "-"). An ad-hoc signature satisfies the CPU requirement but is **not** a Developer ID signature and is not notarised. Unless the package was signed and notarised with a Developer ID by the person who built it, macOS may show a warning the first time.

- **Installer blocked:** Control-click (right-click) the .pkg and choose **Open**, then confirm. Or open **System Settings > Privacy and Security** and press **Open Anyway** next to the blocked item.
- **Plug-in or app blocked after a download:** remove the quarantine flag in Terminal, for example:
  `xattr -dr com.apple.quarantine "/Library/Audio/Plug-Ins/VST3/Phil S Audio/Massive PSXL.vst3"`
- Repeat for the .component and .app if needed. Files you build and install yourself with **install_local.sh** are not quarantined.

First launch

1. **Standalone:** open **Massive PSXL.app** from /Applications/Phil S Audio. Use the standalone window's audio and MIDI settings to choose your interface and keyboard.
2. **In a DAW:** create an instrument track and choose **Massive PSXL** (manufacturer Phil S Audio). Logic Pro lists it as an Audio Unit instrument; other hosts list the VST3.
3. A brand-new instance starts as **Init (Blank)**: the **Routing** tab is open and the synth is silent until you connect something or load a preset. That is expected: see chapter 3.
4. Click **Keys** in the top bar to pull down the on-screen keyboard, or play your MIDI keyboard.

> **TIP.** If a host does not list the plug-in, rescan plug-ins. For Audio Units, Logic runs its own validation on new components; the project also includes a validation step using Apple's **auval** tool (type aumu, subtype Psxl, manufacturer PhSA).

Where Massive PSXL keeps its files

What	Location
User presets	~/Music/Phil S Audio/Massive PSXL/Presets/ (files end in .psxlpreset; presets saved in a category go in a sub-folder)
Performer patterns	~/Music/Phil S Audio/Massive PSXL/Presets/Patterns/ (.psxlperf)
Settings (theme, size, keyboard)	~/Library/Application Support/Phil S Audio/Massive PSXL.settings
Factory presets and artwork	Built into the plug-in; nothing to install separately

Uninstalling

Delete the three items listed above. Your presets and settings files are separate: delete the **Phil S Audio** folders under Music and Application Support as well if you want a clean removal.

3 Quick Start

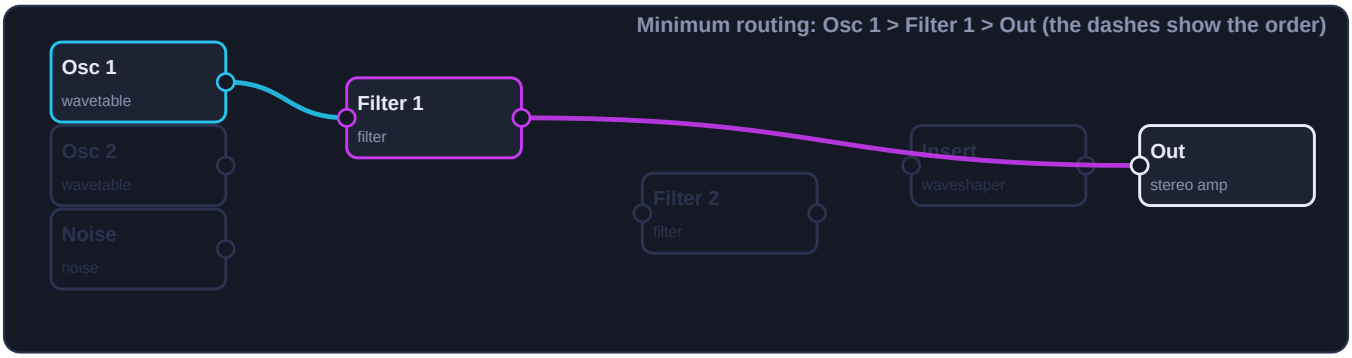
Silence, routing, and your first note in five steps

Massive PSXL has two ways to start: load a ready-made **factory preset**, or build from scratch with **Load Init (Blank)**. This chapter does the second, because it teaches the most important idea in the synth: **nothing is connected until you connect it**.

> **IMPORTANT. Load Init (Blank)** resets every parameter to its default and removes all routing and all modulation. With no routing the synth is **silent** until you connect something in the Routing tab. That is by design, not a fault.

Step by step

1. **Load the blank slate.** Press **Menu** (top bar) and choose **Load Init (Blank)**. The preset name in the top bar becomes **Init (Blank)**. The **Routing** tab shows seven boxes and the message 'No routing connections - the synth is silent.'
2. **Connect Osc 1 to Filter 1.** Put the mouse over the little circle on the *right edge* of the **Osc 1** box (its output port), press and hold, drag to the circle on the *left edge* of **Filter 1** (its input port) and release. A wire appears.
3. **Connect Filter 1 to Out.** Drag from Filter 1's right-edge port to the left-edge port of the **Out** box.
4. **Play.** Click **Keys** in the top bar and click a key, or play your MIDI keyboard. The wire glows while sound passes through it, and the output meters at the top right move.
5. **Shape it.** In the **Osc** panel drag the **Pos** knob and watch the wavetable display scan through its frames. Switch the **Filter 1** panel's **Cutoff** down to make it darker.



The smallest working patch. Everything else is optional.

Shortcut: quick setups

Right-click an **empty spot** in the Routing view to open a menu of quick setups. They clear the routing and wire a common arrangement for you:

- **Osc 1 > Filter 1 > Out**
- **Osc 1 + Osc 2 > Filter 1 > Out**
- **Serial: Osc 1 + 2 > Filter 1 > Filter 2 > Out**
- **Parallel: Osc 1 > Filter 1, Osc 2 > Filter 2 > Out**
- **Osc 1 + 2 + Noise > Insert > Out**
- **Noise > Filter 1 > Insert > Out (drums / FX)**
- **Clear all connections**

Make it move

1. Open the **Mod 2** tab (lower tab strip). Choose Type **LFO**.
2. Drag the **Mod 2** chip (top-left of that tab) onto the **Cutoff** knob of **Filter 1**. The knob grows a coloured ring: the filter is now modulated.
3. Right-click the Cutoff knob and choose **Edit amount: Mod 2** to change how far it sweeps.
4. Drag macro chip **M1** (under the first knob in the macro strip) onto any knob. Macro 1 takes that knob's name automatically.

Or just load a preset

Press **Browse** (next to the preset name) or the preset name itself, pick a category such as **Pads** and click a preset. A single click loads it. Double-click loads it and closes the browser. The **<** and **>** arrows beside the preset name step through the whole library. Factory presets already include their routing, so they sound as soon as you play.

No sound? Checklist

- Is there at least one wire from a source to **Out** (possibly through filters)? Check the Routing tab.
- Is the signal chain complete? A filter or insert with nothing connected into it produces nothing; any wire into Out must come from a node that has input.
- Is the source's **Level** up (Osc Level, Noise Level)? Is **Amp Level** and the master **Volume** above zero?
- Is the amp envelope Sustain and Release sensible? A Decay with Sustain 0 plays only a short blip.
- Is the note reaching the plug-in? The on-screen keys light when MIDI arrives.

4 Interface Tour

Every panel, tab and button

The plug-in window is 1260 by 700 units (850 with the keyboard open) and scales from 75 to 200 percent. It is divided into a **top bar**, a row of **synth panels**, a **lower tab area**, the **macro strip** and an optional **keyboard**.

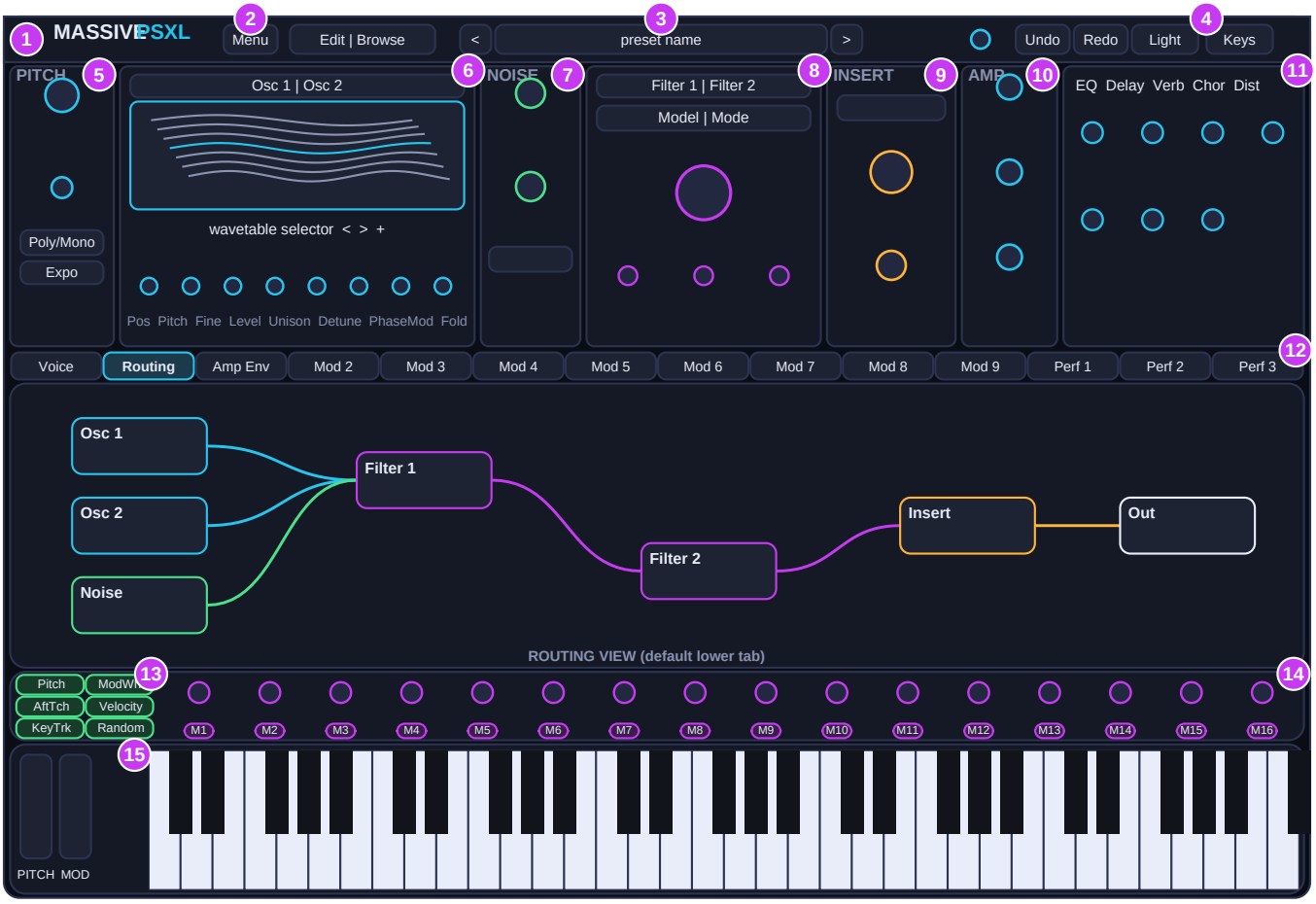


Figure 4.1: Overview of the editor. Numbers refer to the table below.

#	Area	What it does
1	Logo and name	Massive PSXL by Phil S Audio.
2	Menu	Presets, Mutate, Undo/Redo, Settings and About (see 4.2).
3	Edit | Browse tabs, < > and preset name	Switch between the synth and the preset browser; step through presets. Pressing the preset name toggles the browser.
4	Volume, meters, Undo, Redo, Light/Dark, Keys	Master volume knob, stereo output meters, undo history buttons, theme toggle, keyboard drawer.
5	PITCH panel	Global Pitch (+/-48 st), Glide, Voice Mode (Poly / Mono / Legato) and the Expo glide switch.
6	Oscillator panel	Osc 1 | Osc 2 tabs, 3D wavetable display, table selector, + import button and eight knobs.
7	NOISE panel	Level, Color and the noise Type.
8	Filter panel	Filter 1 | Filter 2 tabs, Model and Mode menus, Cutoff, Reso, Drive and KeyTrk.
9	INSERT panel	Insert Type, Drive and Mix.
10	AMP panel	Amp Level, Pan and Feedback.
11	FX panel	EQ, Delay, Verb, Chor and Dist tabs for the master effects.
12	Lower tab strip	Voice, Routing, Amp Env, Mod 2 to Mod 9, Perf 1 to Perf 3.
13	Performance source chips	Pitch, ModWhl, AftTch, Velocity, KeyTrk and Random chips for modulation.
14	Macro strip	16 macro knobs, each with an M1 to M16 assignment chip.
15	Keyboard drawer	Pitch and Mod wheels, octave buttons and the piano keys.

4.1 The top bar

Control	Function
Menu	Opens the main menu (4.2).
Edit | Browse	Two tabs. **Edit** shows the synth. **Browse** opens the preset browser over the synth panels.
< and >	Load the previous or next preset in the full list (factory first, then your own), wrapping around.
Preset name	Shows the current preset. Pressing it toggles the browser.
Volume knob	The master Volume (default 0.7). It is a modulation target for macros, mod wheel and aftertouch.
Meters	Two thin bars (left and right) at the far right; they turn amber near full level.
Undo / Redo	Step back and forward through parameter edits, routing changes, modulation assignments and macro renames. They grey out when empty.
Light / Dark	Toggles the theme. The button is labelled with the theme you will switch to.
Keys	Slides the on-screen keyboard in or out. The setting is remembered.

4.2 The Menu

Menu item	What it does
Load Init (Blank)	Resets every parameter to default, clears routing, modulation and macro names, sets all Performer steps to 0.5 and names the preset 'Init (Blank)'. Silent until you route something.
Save preset...	Opens the Browse tab, where you press **Save preset...** to name and file the sound.
Open user preset folder	Reveals ~/Music/Phil S Audio/Massive PSXL/Presets in Finder (creating it if needed).
Mutate sound (subtle)	Randomly nudges the voice parameters by up to +/-6 percent of each knob's range.
Mutate sound (wild)	The same with up to +/-25 percent. The Pitch and Fine knobs, the oscillator Levels and Amp Level are left alone.
Undo / Redo	Same as the top-bar buttons.
Settings > Dark theme / Light theme	Choose the theme.
Settings > Interface size	75, 100, 125, 150, 175 or 200 percent.
Settings > Keyboard size	2 to 8 octaves for the on-screen keyboard.
Settings > Import wavetable...	Choose an audio file; it becomes a user wavetable on Osc 1.
Settings > Load Scala tuning (.scl)...	Load a microtuning file.
Settings > Reset tuning to 12-TET	Back to 12-tone equal temperament.
About Massive PSXL	Version and credits.

> **NOTE.** Mutate changes only continuous voice parameters (oscillator position, detune, filter, insert, noise and so on). It never changes wavetable choice, filter models, routing or the master effects. Use Undo to return.

4.3 Knobs: the basics

All knobs behave the same way:

- **Drag** up or right to increase, down or left to decrease (about 160 pixels for the full range).
- **Mouse wheel** over a knob nudges it.

- **Hover** over a knob and its label is replaced by the exact value and unit.
- **Right-click** opens a menu with **MIDI Learn**, **Clear MIDI mapping** (when mapped) and **Reset to default**. A modulated knob adds **Edit amount** and **Remove** entries for each modulation source.
- **Drop a modulation chip** on a knob to assign it. A magenta ring appears around a knob that will accept the drop.
- A knob with modulation grows a **coloured arc** showing how far each source can move it, and a **white dot** showing the live value while you play.

4.4 Pitch panel

The leftmost panel holds the global voice settings you reach for most:

Control	Range / choices	Notes
Pitch	-48 to +48 st (continuous)	Transposes everything. A modulation target (the LFO vibrato target).
Glide	0 to 5 s	Portamento time between notes (see chapter 13).
Voice Mode menu	Poly, Mono, Legato	
Expo	on / off	Exponential glide curve.

4.5 Oscillator panel

Choose **Osc 1** or **Osc 2** with the tabs. The 3D **wavetable display** draws the frames of the current table as stacked waves; the highlighted line is the current **Pos**. **Click or drag in the display** to set Pos directly. The row below holds **<** and **>** buttons, the wavetable drop-down and a **+** button to import an audio file as a wavetable (or drop a file on the display). Eight knobs follow: **Pos, Pitch, Fine, Level, Unison, Detune, PhaseMod, Fold**. Chapter 6 describes each.

4.6 Noise, Filter, Insert, Amp and FX panels

- **NOISE**: Level, Color and a Type menu (White, Pink, Brown, Crackle).
- **Filter**: tabs **Filter 1** and **Filter 2**; a **Model** menu (Ladder, State Variable, Comb, Formant) and a **Mode** menu whose entries depend on the model; a large **Cutoff** knob; **Reso**, **Drive** and **KeyTrk**.
- **INSERT**: a Type menu (TanH, Fold, Bitcrush, Hard Clip, Rectify, Sine), **Drive** and **Mix**.
- **AMP**: **Level**, **Pan**, **Feedback**. The amp envelope lives in the Amp Env tab below.
- **FX**: tabs **EQ** (Lo Gain, Lo Freq, Mid Gain, Mid Freq, Mid Q, Hi Gain, Hi Freq), **Delay** (Time, Feedback, Mix, Sync toggle, Division), **Verb** (Size, Damping, Mix), **Chor** (Rate, Depth, Mix) and **Dist** (Drive, Mix).

4.7 The lower tab strip

Fourteen tabs share the large area under the synth panels. The plug-in opens on **Routing**.

Tab	Content
Voice	Voice Mode, Polyphony, Glide, Bend Range, Vel Sens, Expo; a list of imported wavetables; **Import wavetable...** button; microtuning name with **Load Scala (.scl)...** and **12-TET** buttons.
Routing	The node editor (chapter 5).
Amp Env	Envelope display and Delay, Attack, Hold, Decay, Sustain, Release, Env Level, OneShot.
Mod 2 to Mod 9	Eight modulator slots. Each has a Type menu (Envelope or LFO), a source chip, and an envelope or LFO editor.
Perf 1 to Perf 3	Three Performers: 16 draggable step bars, Steps, Rate, Smooth, Sync, Division, **Randomise** and **Patterns**.

4.8 The macro strip and keyboard

The strip along the bottom has two parts. On the left, six **performance-source chips**: **Pitch** (pitch bend), **ModWhl**, **AftTch**, **Velocity**, **KeyTrk** and **Random**. To their right are the **16 macro knobs**, each with a small crosshair chip (**M1** to **M16**) underneath. Drag any chip onto a knob to assign it. **Double-click** a macro chip (or right-click it) to rename the macro.

The **Keys** button pulls the keyboard down below the macro strip. Details are in chapter 15.

4.9 Toasts and feedback

Short messages appear as a coloured pill below the top bar for a few seconds, for example when a modulation is assigned, a file is imported, a preset is saved, or an action is refused (such as connecting a node backwards).

5 Signal Flow and Routing

The node editor and how audio travels

Massive PSXL does not have a fixed signal chain. The **Routing** tab shows seven nodes. You decide which sources feed which processors.

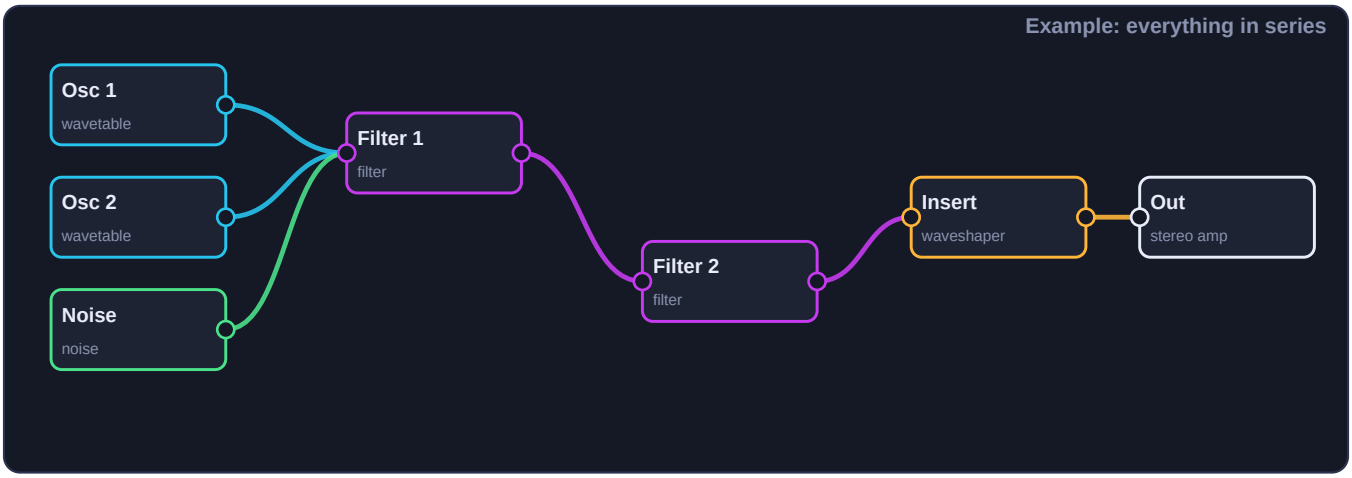


Figure 5.1: Seven nodes: three sources on the left, then Filter 1, Filter 2, Insert and Out.

5.1 The seven nodes

Node	Role	Subtitle shows
Osc 1, Osc 2	Wavetable oscillators.	The selected wavetable name.
Noise	Noise generator.	The noise type.
Filter 1, Filter 2	Two independent filters.	Model / Mode, e.g. 'Ladder / LP 24'.
Insert	Per-voice waveshaper / distortion.	The insert type.
Out	The voice's stereo amp and pan, then the master effects.	'Stereo amp'.

Each node has an **activity meter** (thin bar along the bottom) and ports: an **output port** on the right edge and, for Filter 1, Filter 2, Insert and Out, an **input port** on the left edge.

5.2 The one rule: left to right

Signal can only flow forward. A source may feed any node that comes **later** in this order:

Osc 1, Osc 2, Noise > Filter 1 > Filter 2 > Insert > Out

Output of	Can connect to
Osc 1, Osc 2, Noise	Filter 1, Filter 2, Insert, Out
Filter 1	Filter 2, Insert, Out
Filter 2	Insert, Out
Insert	Out

Trying to connect backwards gives a toast: *Signal flows left to right: Insert can't feed Filter 1*. While you drag a wire, valid input ports light up in amber.

5.3 Making and removing connections

- **Connect**: press on an **output port** (right edge of a node), drag to a compatible **input port** (left edge) and release. The new wire has gain 100 percent.
- **Wires glow with audio**: the thicker and brighter a wire, the more signal is passing through it, and small dots flow along it.
- **Select a wire** by clicking on it (it turns white), then press **Delete** or **Backspace** to remove it.
- **Double-click** a wire to remove it immediately.
- **Right-click a wire** for **Remove connection** and for **Gain 25%, 50%, 75%, 100% or 150%**. A wire that is not at 100 percent shows its gain in a small label at its midpoint.
- **Right-click empty space** for the quick setups (below) and **Clear all connections**.
- Many wires can enter one input: their signals are **summed**. One output can feed several inputs (fan-out).

5.4 Quick setups

Quick setups replace the current routing (and can be undone with **Undo**):

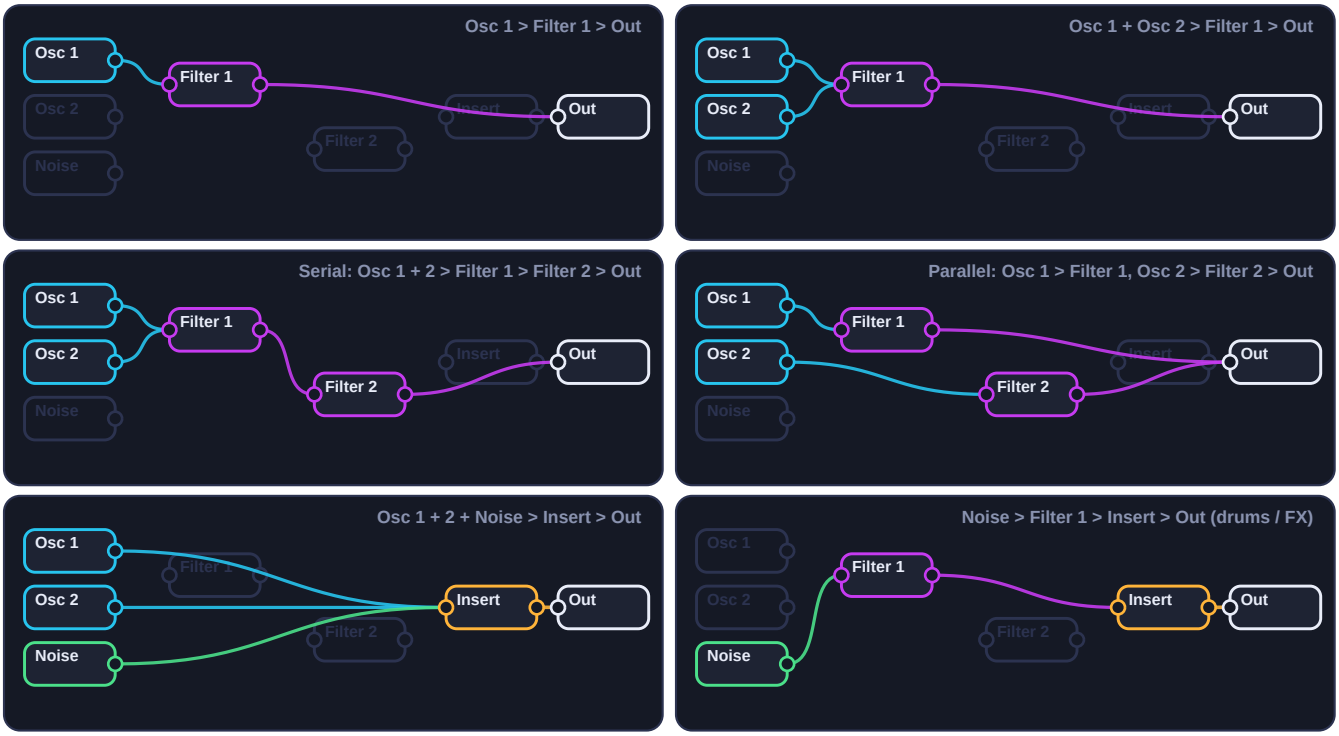


Figure 5.2: The six quick setups available from the right-click menu.

5.5 Things to know about routing

- **Silent by default.** After **Load Init (Blank)** there are no wires. With no wire into **Out** you hear nothing.

- **A processor with nothing connected to it is skipped.** Filter 1, Filter 2 or the Insert do nothing unless a wire enters them, and a wire from them to Out carries silence.
- **Sources always run.** An oscillator that is not wired still runs, so it can act as a **Phase Mod** partner for the other oscillator even though it is silent (see the FM bell tutorial).
- **Series and parallel.** Osc > Filter 1 > Filter 2 is series. Osc 1 > Filter 1 and Osc 2 > Filter 2, both into Out, is parallel.
- **Skipping stages.** Osc 1 can go straight to the Insert or to Out; Noise can bypass the filters; any filter can be bypassed.
- **Gain per wire.** 25 to 150 percent lets you balance sources into a filter or push a filter harder into the Insert.
- **Amp Feedback** (Amp panel) feeds a saturated copy of the voice's output back into **Filter 1**'s input. If Filter 1 has no input it feeds the **Insert** instead.
- Routing is saved in presets and in your DAW project, and is covered by **Undo / Redo**.

> **TIP.** A good habit when building a sound: connect the whole chain first so you can hear it, then refine. The node activity meters and glowing wires tell you where signal is flowing and where it stops.

6 Oscillators and Wavetables

Two scanning oscillators, ten tables, your own audio

A wavetable is a stack of single-cycle waveforms called **frames**. The **Pos** (position) knob chooses which frame you hear and crossfades between neighbouring frames, so turning it morphs the timbre smoothly. Both oscillators have the same controls and can use the same or different tables.

6.1 Oscillator controls

Knob	Range	Default	What it does
Pos	0 to 1	0	Scans through the frames of the wavetable. Click or drag in the 3D display to set it directly.
Pitch	-48 to +48 st	0	Coarse tuning in whole semitones. Under modulation it moves smoothly.
Fine	-100 to +100 ct	0	Fine tuning. Pitch and Fine add together.
Level	0 to 1	0.8	Output level into the routing.
Unison	1 to 8	1	Number of stacked voices per note. Output is scaled by 1/sqrt(n) so the level stays steady.
Detune	0 to 100 ct	15	Spread of the unison stack: voices are spread evenly from -Detune to +Detune cents.
PhaseMod	0 to 1	0	Phase modulation from the *other* oscillator (6.4).
Fold	0 to 1	0	Wavefolder after the oscillator (6.5).

The oscillator tab (Osc 1 or Osc 2) you are looking at decides which oscillator the knobs, the table menu and the display control. Osc 1 defaults to the first table, **Morph Basic**; Osc 2 defaults to the second, **Pulse Sweep**.

6.2 The ten built-in wavetables

Every built-in table has 32 frames. The table below says what you hear as Pos goes from 0 to 1.

#	Wavetable	What Pos does	Good for
1	Morph Basic	Sine (0), triangle (about 0.33), saw (about 0.66), square (1.0), smoothly morphing between them.	Everything: basses, leads, pads, drums.
2	Pulse Sweep	Pulse width from a 50 percent square (0) to a very narrow pulse of about 3 percent (1).	PWM leads and pads, hollow basses, clavs.

#	Wavetable	What Pos does	Good for
3	Harmonic Tilt	Spectral slope from dull and almost sine-like (0) to very bright and buzzy (1).	Keys, soft pads, harp-like plucks.
4	Vowel Choir	Vowels A, E, I, O, U at about 0, 0.25, 0.5, 0.75 and 1. The formants are fixed in the wave shape, so they move with the note you play.	Choirs, vocal leads, talking sounds.
5	Hard Sync	A saw whose sync ratio rises from 1 (plain saw) to 8 (screaming sync).	Sync leads, zaps, bass.
6	FM Bells	A sine with FM that gets brighter and moves through harmonic ratios 1 to 6 as Pos rises.	Bells, electric piano, mallets, FM bass.
7	Drawbar Organ	Morphs through different drawbar-style registrations of harmonics 1, 2, 3, 4, 6, 8, 10, 12 and 16.	Organs, reed-like pads.
8	Glass Harmonics	A bright, odd-harmonic stack with a resonant peak that travels from the low harmonics (3rd) up to about the 43rd.	Glassy pads, bells, shimmering leads.
9	Digital Grit	A random-phase band of harmonics that starts narrow and low and widens and brightens.	Gritty basses, noise-like risers and textures.
10	Fold Sweep	A sine that is progressively wavefolded: 0 is mild, 1 is heavily folded.	Metallic leads and basses.

TIP. Wavetables are band-limited: the engine chooses a version with fewer harmonics at high pitches, so even bright tables stay clean (little aliasing) up the keyboard.

6.3 Unison

Unison stacks up to eight copies of the oscillator, evenly detuned across **Detune**. Each note's unison voices start at random phases (the first voice always starts at phase 0), which gives the lush, never-identical supersaw character. Unison costs CPU: 8 unison voices on 8 notes is 64 oscillators. Reduce **Polyphony** (Voice tab) or Unison if your computer struggles.

6.4 Phase Mod

Each oscillator can phase-modulate the other: **Osc 1's PhaseMod** is driven by Osc 2's output, and **Osc 2's PhaseMod** by Osc 1's. This is the same idea as FM synthesis. The key points:

- Both oscillators always run, even when not wired into the routing or when their Level is 0, so a **silent oscillator can still be the modulator**.
- The modulator's **Pitch** sets the carrier-to-modulator ratio. Whole-number ratios (+12 st = 2:1, +19 st = 3:1, +24 st = 4:1) give harmonic tones; other values give bells and clangs.
- Modulating **PhaseMod** with an envelope makes the tone start bright and mellow out, the classic FM pluck.
- Setting both oscillators' PhaseMod above zero makes them modulate each other (feedback-like FM) and gets rough quickly.

6.5 Fold

Fold pushes the oscillator output into a sine-shaped wavefolder. Low settings add harmonics gently; high settings fold the wave back on itself for metallic, hollow timbres. It reacts to the oscillator's level, so Unison and the table choice matter.

6.6 Wavetable library and importing your own

The wavetable menu shows the ten built-in tables, then your imported tables labelled **User: name**. There are **22 user slots**. Use **<** and **>** beside the menu to step through the loaded tables.

Three ways to import

- **Drag and drop** a WAV, AIFF, FLAC or OGG file onto the wavetable display. It loads into the oscillator whose tab is open. The display's frame lights up magenta while you hover.
- Press the **+** button next to the wavetable menu in the Oscillator panel (also loads into the open oscillator).
- **Menu > Settings > Import wavetable...** or the **Import wavetable...** button in the **Voice** tab (these load into Osc 1).

How a file becomes a wavetable

- The file is mixed down to mono.
- It is divided into frames: the number of frames is the length divided by 2048 samples, rounded, and limited to 1 to 256. Each frame is resampled to 2048 points.
- A short single-cycle file (say 2048 samples) therefore becomes **one frame**; a long recording becomes a many-frame table you can scan with Pos.
- Each frame has its DC offset removed and is peak-normalised, then band-limited like the built-in tables.
- The new table is named after the file, placed in the first free user slot (when all 22 are full the last slot is replaced) and selected automatically.
- Imported tables are saved with the DAW session and with **user presets**, so presets that use them travel with the table data. Factory presets only use built-in tables.
- If the file cannot be read, the toast says *Could not read that audio file*.

TIP. For the cleanest results use single-cycle waveforms of exactly 2048 samples, or several of them joined end to end (for example 64 cycles = 131072 samples). Recordings of voices, instruments or field sounds work too: they become evolving textures.

7 Noise

White, pink, brown and crackle

The **NOISE** panel is a third source. Like the oscillators it does nothing until you wire the **Noise** node in the Routing tab.

Control	Range / choices	Default	What it does
Level	0 to 1	0.5	Output level into the routing.
Color	0 to 1	0.5	A tone control: below 0.5 a low-pass (darker, from 20 kHz down to about 100 Hz at 0), 0.5 is neutral (unfiltered), above 0.5 a high-pass (thinner, up to about 10 kHz at 1).
Type	White, Pink, Brown, Crackle	White	Noise colour (spectrum) or texture.

Noise types

Type	Character	Typical use
White	Equal energy at every frequency: bright, hissy.	Hats, snares, risers, breath.
Pink	Energy falls 3 dB per octave: balanced, natural.	Wind, rain, softer percussion.
Brown	Energy falls 6 dB per octave: deep rumble.	Ocean, thunder, low drones.
Crackle	Sparse random clicks that decay quickly, like vinyl dust.	Texture, lo-fi crackle, metallic hits through a Comb filter.

TIP. Noise is a modulation target: you can modulate its **Level** and **Color**. A short envelope on Noise Level gives a click on a kick; a slow LFO on Color gives breathing, swooshing noise.

8 Filters

Four models, sixteen modes, two filters

Each of the two filters has a **Model** and a **Mode**. Changing the Model changes the list of Modes. Controls common to all models:

Control	Range	Default	Notes
Cutoff	20 Hz to 20 kHz	F1 12000 Hz, F2 8000 Hz	Mid-travel is 1 kHz (a skewed scale, fine control at low frequencies). In Comb it sets the comb's pitch; in Formant it picks the vowel.
Reso	0 to 1	0.15	Resonance. Its effect depends on the model.
Drive	0 to 1	0	Saturation *before* the filter (gain from 1x up to 8x, with level compensation at low signals).
KeyTrk	-1 to +1	0	Key tracking around note 60. +1 makes the cutoff follow the keyboard 1:1 (an octave per octave); negative values open the filter on low notes.

8.1 Ladder

A four-stage transistor-ladder-style low-pass with a soft saturating feedback path. Warm, fat and the best for basses and acid lines. Near the top of the Reso range it rings almost like a sine.

Mode	Result	Slope
LP 24	Four-pole low-pass: the classic fat filter.	24 dB/octave
LP 12	Two-pole low-pass: brighter, more open.	12 dB/octave
BP	Band-pass between the second and fourth stages: a narrow, nasal band.	roughly 12 dB each side
HP 24	Four-pole high-pass: removes lows, good for thinning.	24 dB/octave

8.2 State Variable

A clean 12 dB/octave multimode filter. Resonance is precise and musical rather than aggressive; Reso 1 is strongly resonant but stable.

Mode	Result
LP	Low-pass.
BP	Band-pass: the choice for snares, claps and telephone tones.
HP	High-pass: hats and shakers.
Notch	Removes a band: phaser-like when swept.

8.3 Comb

A short feedback delay line. **Cutoff** sets the delay time as a frequency: the comb resonates at the Cutoff frequency and its harmonics. Set **KeyTrk** to +1 and tune **Cutoff** so the comb follows the keys. **Reso** controls the feedback.

Mode	Result
Comb +	Positive feedback: bright ringing at the comb frequency and all its harmonics.
Comb -	Negative feedback: hollow, odd-harmonic resonance.
Flange	Feed-forward comb: moving notches (jet-plane flanging when Cutoff is modulated). Reso has no effect in this mode.

Mode	Result
Pluck	Damped feedback: a Karplus-Strong style plucked-string resonance. Reso sets how long the string rings. Works with a noise or short sound as the exciter.

8.4 Formant

Three resonant band-pass filters tuned to vocal formants. **Cutoff** no longer means a frequency of cut-off: it **scans through the vowels** A, E, I, O, U. **Reso** sets the sharpness of the formants (from soft to very vowel-like).

Cutoff (approx.)	Vowel
20 Hz (fully anticlockwise)	A
about 110 Hz	E
about 630 Hz	I
about 3.5 kHz	O
20 kHz (fully clockwise)	U (reached at the very end)

Mode	Voice type
Male	Adult male formant frequencies.
Female	Adult female formant frequencies (higher).
Child	The male set scaled up by 35 percent.
Alien	The male set scaled down by 30 percent with the second formant shifted up by 50 percent.

Between the vowel points the formants interpolate smoothly, so sweeping the cutoff with an envelope or LFO makes the sound speak.

8.5 Using two filters

- **Series** (Filter 1 > Filter 2): steeper or combined responses, e.g. Ladder LP 24 into Formant for a vocal bass.
- **Parallel** (Osc 1 > Filter 1, Osc 2 > Filter 2, both to Out): each oscillator gets its own filter, e.g. a low-passed sub plus a high-passed layer.
- **Split roles**: one filter for tone (Ladder), the other for character (Comb or Formant).
- Filter 2 has its own Cutoff, Reso, Drive, KeyTrk, Model and Mode; its default Cutoff is 8000 Hz (Filter 1: 12000 Hz).

> **TIP.** **Filter Drive** and **Amp Feedback** are different: Drive saturates the signal entering a filter; Feedback loops the whole voice's output back to Filter 1. Both thicken the sound.

9 Insert and Amp

Per-voice shaping, level, pan and feedback

9.1 The Insert

The **INSERT** is a waveshaping effect that sits inside each voice. It works like a node: wire a source or filter into it and wire it onward to Out (or nothing hears it). **Drive** (0 to 1, default 0.3) sets how hard the signal is pushed (about 1x to 15x gain) and **Mix** (0 to 1, default 0.5) blends dry and shaped signal. Both are modulation targets.

Type	What it does	Sound
TanH	Smooth soft saturation.	Warm analogue-style distortion; keeps low end tidy.
Fold	Sine-shaped wavefolder.	Metallic, hollow, rich overtones; reacts strongly to level.
Bitcrush	Drive lowers the bit depth (from about 15 bits down to about 3) and holds each sample for up to 13 samples (reduces sample rate).	Lo-fi, digital grit, aliasing.
Hard Clip	Flat-tops the waveform at full scale.	Aggressive, square-ish distortion.
Rectify	Full-wave rectification (flips negative halves).	Octave-up buzz and sub-bass reinforcement.
Sine	A sine waveshaper that folds back at high drive.	Smooth at low drive, glassy at high drive.

TIP. Drums: *Noise* > *Filter 1* > *Insert* > *Out* is the quick setup for hats and snares. A fold or TanH insert with Mix 1.0 thickens a kick's attack.

9.2 The Amp

Every voice ends in the amp. Its controls are:

Control	Range	Default	What it does
Level	0 to 1	0.7	Per-voice level after the amp envelope. A modulation target. Setting it to 0 and letting a Performer or LFO raise it is how gates work.
Pan	-1 to +1	0	Equal-power stereo position of the voice. Modulate it for auto-pan.
Feedback	0 to 1	0	Feeds a soft-saturated copy of the voice's output back into the input of Filter 1 (or the Insert if Filter 1 is not in use). Adds grit, resonance and instability.

The amp envelope is the **Amp Env** tab (chapter 11). The voice level is: amp envelope x Amp Level x velocity scale, where the velocity scale is (1 - Vel Sens) + Vel Sens x velocity. With Vel Sens 0 every note plays at full level; with 1 the level follows your playing.

NOTE. The master **Volume** (top bar) is separate and comes after the effects. Its curve is volume squared (times 2), so 0.7 is close to unity gain and 1.0 is about +6 dB. A soft safety limiter smooths peaks above about 0.95 (full scale).

10 Master Effects

EQ, distortion, chorus, delay and reverb

All voices are summed and sent through five stereo effects, always in this order, and finally through the master Volume. Each effect is bypassed when its Mix is 0 (the EQ when all three gains are about 0 dB), so unused effects cost no CPU.

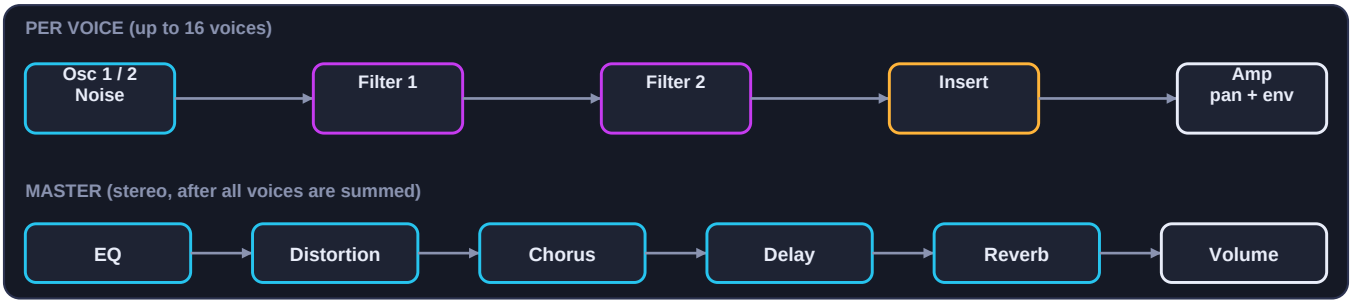


Figure 10.1: Voice processing happens per note; the master effects process the summed stereo signal.

A DC blocker runs ahead of the EQ to remove any constant offset (for example from the Rectify insert). The effects are controlled in the **FX** panel on the right of the synth row, with five tabs.

10.1 EQ

A three-band equaliser: a low shelf, one fully parametric mid band and a high shelf.

Control	Range	Default
Lo Gain / Lo Freq	-18 to +18 dB / 40 to 400 Hz	0 dB / 120 Hz
Mid Gain / Mid Freq / Mid Q	-18 to +18 dB / 200 Hz to 8 kHz / 0.3 to 6	0 dB / 1000 Hz / 1.0
Hi Gain / Hi Freq	-18 to +18 dB / 2 to 16 kHz	0 dB / 6000 Hz

Modulatable: Lo Gain, Mid Gain, Mid Freq and Hi Gain (by macros, mod wheel, aftertouch or pitch bend).

10.2 Distortion

Dist tab: **Drive** (0 to 1, default 0.3) is a tanh saturator with gain from 1x to 13x; **Mix** (0 to 1, default 0 = off) blends it with the dry signal. Because it comes after the EQ and before chorus, delay and reverb, it distorts the whole mix but not the effects tails.

10.3 Chorus

Chor tab: a stereo chorus with two short modulated delay lines (left and right a quarter cycle apart). **Rate** 0.05 to 8 Hz (default 0.6), **Depth** 0 to 1 (default 0.5), **Mix** 0 to 1 (default 0). Good for width on pads and basses.

10.4 Delay

Delay tab: a stereo delay with the left and right feedback paths crossed, and a low-pass in the feedback loop so repeats get darker.

Control	Range	Default	Notes
Time	1 to 2000 ms	350 ms	Skewed: 300 ms at mid-travel. Changes glide rather than click.
Sync	on / off	off	When on, the delay time follows the host tempo and **Division**.
Division	1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T	1/8	Only used with Sync on. (T = triplet.)

Control	Range	Default	Notes
Feedback	0 to 0.95	0.35	Modulatable.
Mix	0 to 1	0	Wet level added on top of the dry signal. Modulatable.

At 120 BPM (also used when the host gives no tempo, as in the standalone app) a 1/8 note delay is 250 ms. The longest delay is about 2 seconds; at tempos slower than 120 BPM the longest divisions are limited to that.

10.5 Reverb

Verb tab: a stereo algorithmic reverb. **Size** (0 to 1, default 0.5) sets room size and decay, **Damping** (0 to 1, default 0.5) removes highs from the tail, **Mix** (0 to 1, default 0) is the amount. Mix raises the wet level and slightly lowers the dry. The plug-in reports a 4 second tail to the host.

10.6 Master Volume and safety

Volume (top bar, default 0.7) is the last stage, followed by a soft clip near full scale to keep stray peaks tidy. Volume is a modulation target for macros, mod wheel and aftertouch.

> **NOTE.** Voice modulators (envelopes, LFOs, Performers, Velocity, Key Track, Random) are different for every note, so they cannot control these shared master effects. Use **macros, the mod wheel, aftertouch or pitch bend** to modulate effect parameters (chapter 11).

11 Modulation

Envelopes, LFOs, Performers, macros and controllers

Modulation is anything that moves a knob for you while you play. Every modulation is an **assignment**: a **source** controlling a **destination** knob by an **amount**. You can have up to 256 assignments, several sources on one knob, and one source on many knobs.

11.1 How modulation adds up

The destination's **final value** = knob position + (amount x source value) for every assignment, limited to the knob's range. All of it is done on the knob's own 0 to 1 travel, so the amount is a **fraction of the knob's full travel**:

- An amount of **+0.50** lets the source push the knob half of its travel upward at the source's maximum; **-0.50** pulls it downward.
- Because Cutoff has a skewed scale, +0.4 from a 1 kHz knob position (halfway) reaches about 12.6 kHz at the source's peak.
- For a Pitch knob (range 96 st from -48 to +48) an amount of **0.125 = 12 semitones**, 0.25 = 24 st, and 0.003 is a gentle 0.3 st vibrato.
- Bipolar sources (LFO, Key Track, Random, Pitch Bend) move the value both above and below the knob; unipolar ones (envelopes, Performers, macros, wheels) move it one way from the knob position.

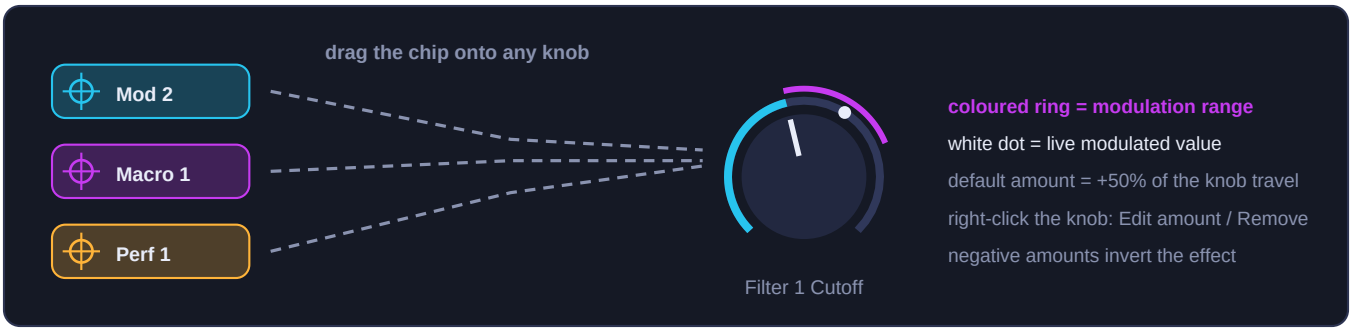


Figure 11.1: Dragging any modulation chip onto a knob creates an assignment.

11.2 Sources

Source	Value range	Scope	Notes
Amp Env	0 to 1 (x Level)	Voice	Slot 1, always an envelope. Its chip is in the Amp Env tab.
Mod 2 to Mod 9	Envelope 0 to 1 (x Level); LFO -1 to +1 (or 0 to +1 when Bipolar is off), x Level	Voice	Eight slots; each is an envelope or an LFO.
Perf 1 to Perf 3	0 to 1	Voice	The current step's bar height.
Macro 1 to Macro 16	0 to 1	Global	The macro knobs. Can control anything, including master effects.
Velocity	0 to 1	Voice	Key velocity of the note.
Key Track	-1 to +1	Voice	(note - 60) / 60, so middle C = 0, MIDI note 0 = -1 and 120 = +1.
Random	-1 to +1	Voice	A new random value for each note.
Mod Wheel	0 to 1	Global	MIDI CC 1 (or the on-screen Mod wheel).
Aftertouch	0 to 1	Global	MIDI channel pressure.
Pitch Bend	-1 to +1	Global	The pitch wheel position (independent of Bend Range, which only controls actual pitch).

IMPORTANT. Voice sources cannot control the master effects. The effects process the combined signal of all notes, so a per-note envelope or LFO has no single value to apply. Dropping one on a master-effect knob shows the toast: *Voice modulators can't control global effects. Use a macro, mod wheel or aftertouch.*

11.3 Envelopes

Slot 1 (**Amp Env**) is always an envelope and controls the loudness of every note. **Mod 2 to Mod 9** can be envelopes too (Type: Envelope). All of them have the same stages:

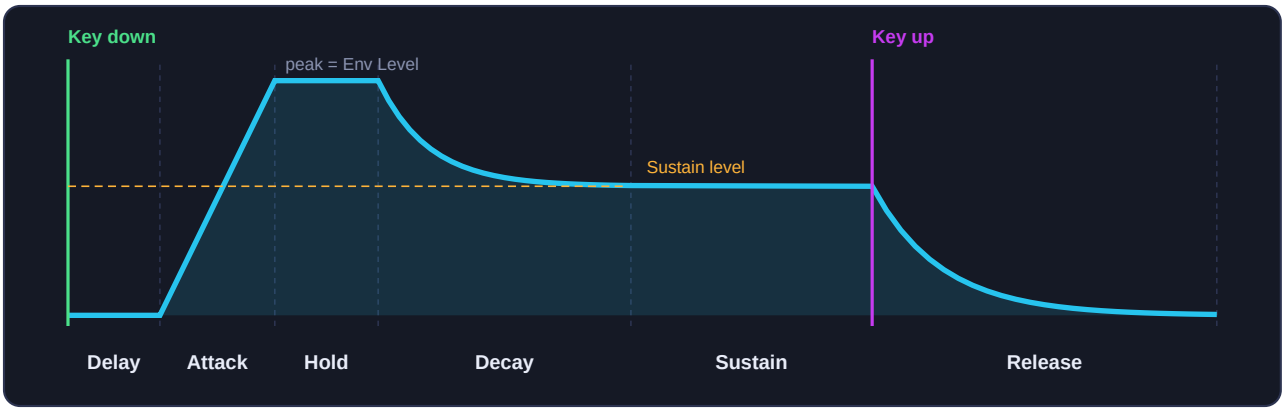


Figure 11.2: Delay, Attack, Hold, Decay, Sustain (while the key is down) and Release (after key up).

Knob	Range	Default (Amp Env / Mod)	Meaning
Delay	0 to 5 s	0	Wait after key down before the attack begins.
Attack	0.5 ms to 10 s	5 ms / 10 ms	Linear rise to the peak.
Hold	0 to 5 s	0	Time spent at the peak before decay.
Decay	1 ms to 10 s	0.4 s	Fall toward the Sustain level (exponential).
Sustain	0 to 1	0.8 / 0.5	Level held while the key is down.
Release	1 ms to 10 s	0.25 s / 0.3 s	Fall to zero after key up (exponential).
Env Level	0 to 1	1	Scales the whole envelope's output (the 'Level' parameter).
OneShot	on / off	off	Ignores key release: the envelope plays through its stages and releases by itself.

- Decay and Release are exponential: they get (practically) all the way to their target in the time shown, which sounds natural for percussion and plucks.
- A new attack starts from the envelope's current value, so retriggering never clicks.
- **OneShot** is the key to drums: with OneShot on, a short tap plays the whole decay. For percussion also set Sustain to 0.
- **Drag the three handles** on the envelope display: the peak point sets Attack, the middle point sets Decay (horizontal) and Sustain (vertical), the end point sets Release. A dot shows the live position of the newest note.
- The voice (and its CPU use) ends when the Amp Env has finished its release.

11.4 LFOs

Choose **Type: LFO** in Mod 2 to Mod 9 to make the slot an LFO. LFOs are **per voice**: each note starts its own LFO at note-on.



Figure 11.3: The seven LFO shapes.

Control	Range	Default	Meaning
Shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine	Waveform. Sample & Hold picks a new random level each cycle; Smooth Random glides between them.

Control	Range	Default	Meaning
Rate	0.01 to 50 Hz	2 Hz	Speed (when Sync is off).
Sync / Division	off, on / 1/1 to 1/16T	off / 1/4	With Sync on, the Rate knob is replaced by a Division menu and the LFO follows host tempo (120 BPM if none).
Phase	0 to 1	0	Where in its cycle the LFO starts at each note-on.
Fade In	0 to 10 s	0	Fades the LFO in after note-on (delayed vibrato).
Level	0 to 1	1	Output depth.
Bipolar	on / off	on	On: swings -1 to +1 around the knob. Off: only 0 to +1 (it never pulls the knob below its setting).

11.5 Performers

A **Performer** is a 16-step sequencer that outputs a value between 0 and 1 per step. There are three. Draw the steps in the **Perf 1, 2, 3** tabs, then drag the Performer's chip onto any knob.

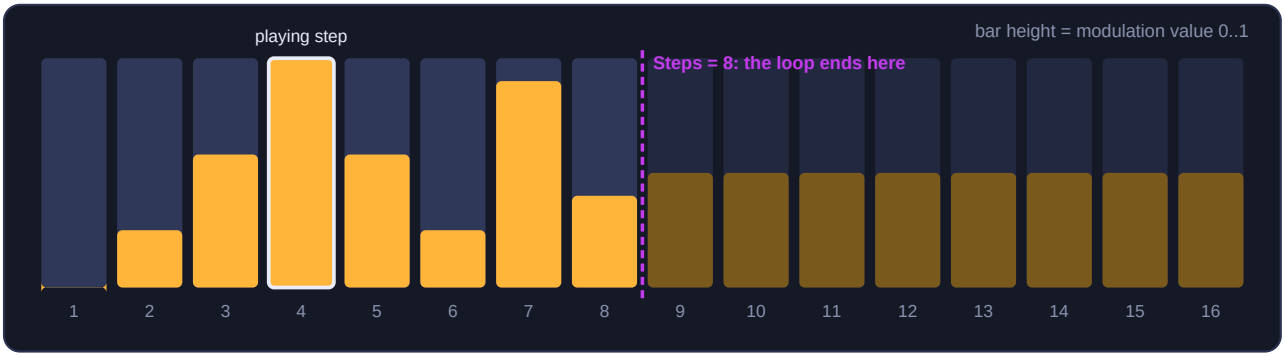


Figure 11.4: Steps 1 to 8 loop (Steps = 8). The outlined bar is the step currently playing.

Control	Range	Default	Meaning
Steps	1 to 16	8	Loop length. Bars beyond Steps are dimmed.
Sync / Division	on, off / 1/1 to 1/16T	on / 1/16	With Sync on, one step lasts one Division at the host tempo.
Rate	0.1 to 40 Hz	4 Hz	Steps per second when Sync is off.
Smooth	0 to 1	0	Glide between steps; 1 glides across the whole step, 0 jumps.
Randomise	button		Fills the bars with random values.
Patterns	menu		Ramp up, Ramp down, Triangle, Alternate high / low, Random, Flat (centre), All zero, Gate: on/off pairs; Save pattern...; Load pattern...

- **Click or drag across the bars** to draw values. Bar height is the modulation value.
- Each note has its own Performer position that restarts at **note-on**, so chords played together stay in step.
- The steps are shared by all notes and saved in presets.
- **Save pattern...** writes a small text file (.psxlperf, sixteen comma-separated values between 0 and 1) to the Presets/Patterns folder. **Load pattern...** reads one back. You can write exact values yourself in a text editor: this is the

precise way to program pitch arpeggios.

- Typical uses: pitch (arpeggio), Amp Level (gate), Filter 1 Cutoff (stepped filter), Osc Position (stepping timbre), Pan, Insert Mix.

11.6 Macros and performance sources

The 16 **macros** are knobs in the bottom strip. A macro does nothing by itself: assign it to one or more knobs and it moves them all. Macros are global, so they can control master effects. They are normal host-automatable parameters, and can be MIDI-learned. Chapter 12 explains automatic naming.

The six chips at the left of the macro strip are the performance sources: **Pitch** (pitch bend), **ModWhl**, **AftTch**, **Velocity**, **KeyTrk** and **Random**. Their chips are green; Mod chips are cyan, Performer chips amber and macro chips magenta.

11.7 Assigning, editing and removing

1. Make the destination knob visible (for example open the Filter 1 tab, or the Osc 2 tab).
2. Press and drag a **chip**: a crosshair appears under the pointer. Chips live in: the top-left of each Mod tab, the Amp Env tab, each Perf tab, and the macro strip.
3. Drop it on the knob. A toast confirms: *Mod 2 now controls Filter 1 Cutoff*. The default amount is **+0.50**.
4. To change the amount, **right-click the knob** and choose **Edit amount: Mod 2 (50%)**. A small slider from -1 to +1 appears; drag it and listen. Negative values invert the effect.
5. To remove, right-click the knob and choose **Remove Mod 2**.
6. Dropping the same chip on the same knob again simply resets its amount to 0.5.

- A knob with modulation shows a coloured **ring** in the source's colour: it is the range the source can reach. A **white dot** shows the live modulated value.
- A source chip is brighter when it is in use.
- Assignments, amounts and macro names are saved in presets and your DAW project, and are covered by Undo / Redo (editing the amount slider is applied live and not stored as separate undo steps).
- Not every knob can be modulated: see the destination list below. Switches, menus, Unison, the wavetable choice, filter Model/Mode, KeyTrk, envelope and LFO settings themselves cannot be modulated.

11.8 Modulation destinations (42)

Voice destinations (28) can be controlled by any source:

Section	Destinations
Osc 1	Position, Pitch, Fine, Level, Detune, Phase Mod, Fold
Osc 2	Position, Pitch, Fine, Level, Detune, Phase Mod, Fold
Noise	Level, Color
Filter 1	Cutoff, Reso, Drive
Filter 2	Cutoff, Reso, Drive
Insert and Amp	Insert Drive, Insert Mix, Amp Level, Amp Pan, Amp Feedback
Voice	Pitch (the global Pitch knob)

Global destinations (14) are the master effects and volume. Only macros, Mod Wheel, Aftertouch and Pitch Bend can control them:

Section	Destinations
Master	Volume
EQ	Low Gain, Mid Freq, Mid Gain, High Gain

Section	Destinations
Delay	Feedback, Mix
Reverb	Size, Mix
Chorus	Rate, Depth, Mix
Distortion	Drive, Mix

12 Smart Macro Naming

Macros take the name of what they control

When you drag a macro chip onto a knob, the macro is automatically **renamed after the knob**. The label under the macro knob changes from 'Macro 1' to, for example, 'Filter 1 Cutoff'. This also changes the **parameter name that the plug-in reports to your host**. Controllers and hardware that display parameter names (see the note below) therefore show a meaningful label instead of 'Macro 1'.

12.1 The rules

- A macro takes the **full parameter name of the first knob** it is dropped on, such as 'Filter 1 Cutoff', 'Osc 1 Position', 'Reverb Mix' or 'Amp Feedback'. The toast reads: *Macro 1 now controls Filter 1 Cutoff (macro renamed to "Filter 1 Cutoff")*.
- Dropping the same macro on more knobs **keeps the first name**, so a macro that controls Dist Drive and Dist Mix stays 'Dist Drive'.
- If you **remove the macro's last assignment** and you never renamed it by hand, the name returns to 'Macro n'.
- If you **rename a macro yourself** (double-click its chip, or right-click the chip and choose **Rename macro...**), your name is kept until you choose **Reset name**. Typing an empty name also resets it.
- Names are limited to 32 characters when generated and are saved in presets and in the DAW project.
- Renaming and assignment are undoable.

12.2 Renaming by hand

Double-click a macro chip (the small crosshair under the macro knob). A dialog titled **Rename Macro n** appears with a text box and the note that the name is also shown on hardware displays. Press **OK** to keep it. Right-clicking a chip gives **Rename macro...** and **Reset name**.

12.3 Hardware and host displays

The macro parameters (Macro 1 to 16) come first in the plug-in's parameter list, in two groups of eight: **Macros 1-8** and **Macros 9-16**. That means a controller that shows parameters eight at a time (an eight-knob page layout, a host's generic controller mapping, or an automation list) shows the macros first. When a name changes, Massive PSXL tells the host that the parameter information has changed; hosts that react to that notification refresh the labels straight away. How quickly or whether a particular host or controller updates its display depends on that host.

> **TIP.** Design a patch so its eight most useful controls are Macros 1 to 8 and give each a sensible name. They will then also be the first page on a hardware controller. Most factory synth presets map Macro 1 to Filter 1 Cutoff, Macro 2 to Filter 1 Reso, Macro 3 to Osc 1 Position and Macro 4 to Reverb Mix.

12.4 Macros and MIDI

Right-click a macro knob and choose **MIDI Learn**, then move a hardware knob or fader: the CC is now tied to that macro (chapter 15).

13 Voice, Glide and Tuning

Polyphony, mono modes, portamento and microtuning

Voice settings are in the **PITCH** panel (Pitch, Glide, Voice Mode, Expo) and in the **Voice** lower tab (Voice Mode, Polyphony, Glide, Bend Range, Vel Sens, Expo, wavetable library and tuning).

Control	Range / choices	Default	Notes
Voice Mode	Poly, Mono, Legato	Poly	
Polyphony	1 to 16	8	Maximum simultaneous notes in Poly mode.
Glide	0 to 5 s	0 (off)	Time to slide from the previous note to the new one.
Expo	on / off	off	Exponential glide curve (fast at first, slowing down).
Bend Range	0 to 24 st	2	How far the pitch wheel bends.
Vel Sens	0 to 1	0.5	How much key velocity affects loudness.
Pitch	-48 to +48 st	0	Global transpose, continuous.

13.1 Voice modes

- **Poly**: up to Polyphony notes at once. When all voices are in use, the oldest note that has been released is stolen first, then the oldest sounding note.
- **Mono**: a single voice. Every new note re-triggers the envelopes. Releasing a key while others are held jumps back to the last held note.
- **Legato**: like Mono, but a note played while another is held **changes pitch without retriggering** the envelopes and Performers. Detached notes retrigger normally. Together with Glide this gives slides (acid lines, expressive leads).
- Switching from Poly to Mono or Legato releases the extra voices.

13.2 Glide

With **Glide** above 0, a new note starts from the pitch of the previous note and slides to the new pitch. In the default (linear) curve the slide always takes the Glide time whatever the interval. With **Expo** on the slide is exponential: it starts fast and settles in about the Glide time. Glide also works in Poly mode (each new note slides from the last played note).

13.3 Pitch, bend and sustain

Pitch wheel moves pitch by up to **Bend Range** semitones (set it to 0 to ignore the wheel for pitch while still using Pitch Bend as a modulation source). The **sustain pedal** (CC 64) holds released notes until the pedal comes up. **All Notes Off** releases every voice; **All Sound Off** silences them at once.

13.4 Microtuning with Scala files

Massive PSXL can play any scale defined in a standard Scala **.scl** file.

1. Open the **Voice** tab and press **Load Scala (.scl)...** (or use **Menu > Settings > Load Scala tuning (.scl)...**).
2. Choose a .scl file. The toast says 'Tuning loaded: name' and the Voice tab shows **MICROTUNING: name**.
3. The scale repeats every period (usually the octave). Degree 0 of the scale sits on MIDI note 60 (middle C, 261.6256 Hz).
4. **12-TET** (Voice tab) or **Menu > Settings > Reset tuning to 12-TET** restores equal temperament. A file that is not a valid scale shows 'Not a valid .scl file'.

NOTE. The tuning is saved with your DAW project (the plug-in's state) and is not changed by loading presets. Only .scl scale files are supported (not .kbm keyboard maps). Ratios, cents and the usual Scala formats are accepted.

14 Presets and the Browser

117 factory sounds, your own library

Click **Browse** (top bar) or the preset name to open the browser over the synth panels. Press **Edit** (or **Close**) to return.

14.1 Browsing

Part	What it does
Category list (left)	All, Bass, Leads, Pads, Plucks, Keys, Drums, FX, Sequence, User.
Search box	Type part of a preset name, category or author to filter the current list.
Preset list	Name, category and author. A **single click loads** the preset (the current one is highlighted). A **double-click** or Return loads it and closes the browser.
Save preset...	Names and saves your sound (14.3).
Open user folder	Opens your presets folder in Finder.
Close	Returns to the synth.
Header	'PRESET BROWSER - 117 factory presets'.

The **<** and **>** arrows in the top bar step through the whole library in browser order (category, then name; your presets follow the factory ones) without opening the browser.

14.2 The factory library

All 117 presets were written for Massive PSXL. Each loads with its own routing, modulation and macro names. See Appendix A for the complete list.

Category	Presets	Flavour
Bass	20	Subs, wobbles, reese, acid, FM, formant and distorted basses.
Leads	19	Saw, supersaw, PWM, sync, FM, glass, vowel, brass and expressive mono leads.
Pads	20	Strings, glass, choirs, evolving and ambient pads.
Plucks	15	Plucks, bells, marimba, kalimba, koto, steel drum and stabs.
Keys	10	E-pianos, organs, clav, digital piano.
Drums	15	Kicks, snares, hats, clap, shaker, rim, toms, cowbell. Play around C0 to C3. Their amp envelopes are OneShot, so a tap plays the whole hit.
FX	10	Risers, falls, zaps, textures, drones and hits.
Sequence	8	Arpeggios and gates built with Performers. Hold a note or chord.

14.3 Saving your own presets

1. Choose **Menu > Save preset...** (it opens the Browse tab) or open the browser yourself.
2. Press **Save preset...**. A dialog asks for a **name** (a fresh Init patch suggests 'My Preset') and a **category** (Bass, Leads, Pads, Plucks, Keys, Drums, FX, Sequence or User).
3. Press **Save**. The toast says *Saved preset "name"*.

4. The file is stored in **~/Music/Phil S Audio/Massive PSXL/Presets**. A preset saved in a category goes in a sub-folder with that name; **User** saves in the main folder. Files end in **.psxlpreset**.

- A user preset stores every parameter, the routing, all modulation assignments with their amounts, macro names, the three Performer patterns and any imported wavetables.
- Your presets appear under **User** (and in search). Copy a .psxlpreset file to another Mac to share it. The browser rescans your folder every time it opens.
- Saving with an existing name overwrites that file.
- **Undo** history is cleared whenever a preset loads.

14.4 Init and Mutate

Menu > Load Init (Blank) gives a completely blank slate: defaults for everything, **no routing**, no modulation, macros named 'Macro 1' to 'Macro 16', Performer bars at 0.5. **Menu > Mutate sound (subtle / wild)** randomly nudges the continuous voice parameters. Use it to explore around a sound; **Undo** reverts it.

14.5 Presets, programs and your DAW

The plug-in presents its 117 factory presets to the host as **programs**, so hosts with a program or preset menu can select them. Your own presets are loaded from the Massive PSXL browser. The full sound (including routing, modulation, macro names, Performer bars, user wavetables, tuning and MIDI-learn mappings) is saved inside your DAW project, so a session reopens exactly as you left it.

15 MIDI, Keyboard and MIDI Learn

Playing, controllers and mapping

15.1 MIDI messages understood

Message	Effect
Note on / off	Plays notes. Velocity controls loudness (Vel Sens) and is available as a modulation source. All MIDI channels are accepted.
Pitch bend	Bends pitch by Bend Range; also the Pitch Bend modulation source.
CC 1 (mod wheel)	The Mod Wheel modulation source.
CC 64 (sustain)	Sustain pedal (value 64 or more = down).
Channel pressure	The Aftertouch modulation source.
All Notes Off / All Sound Off	Releases or silences all voices.
Other CCs	Can be mapped to any parameter with MIDI Learn.

15.2 The on-screen keyboard

Press **Keys** in the top bar. The window grows 150 units taller and a keyboard slides in under the macro strip. Press **Keys** again to hide it. The setting is remembered.

- **Pitch wheel** (left): drag up or down; it springs back to the centre when released.
- **Mod wheel** (next to it): stays where you leave it. Both wheels also follow incoming MIDI from your hardware.
- The small **<** and **>** buttons shift the keyboard down or up by one octave; the label below them shows the lowest octave (C0 at the start; middle C is C3).
- **Menu > Settings > Keyboard size** sets 2 to 8 octaves (default 5).

- The height at which you click a key sets the velocity (lower on the key = harder).
- **Keys light up when MIDI arrives** from a hardware keyboard or the DAW, so you can see what the plug-in receives.
- With the keyboard focused the computer's typing keys also play notes using the standard on-screen-keyboard mapping (A, W, S, E, D, F, T, G, Y, H, U, J, K...).

15.3 MIDI Learn

Any **knob** (including the macro knobs) can be tied to a MIDI controller. Switches and menus are not learnable.

1. **Right-click the control** and choose **MIDI Learn**. The control gets an amber ring and the menu entry reads 'Waiting for MIDI CC... (click to cancel)'.
2. Move a knob, fader or button on your controller. The first CC received becomes the mapping.
3. The control now follows that CC (value 0 to 127 spans the control's range). Right-click it to see **Clear MIDI mapping (CC n)**.

- CC 1 (mod wheel), CC 64 (sustain) and the channel-mode messages (CC 120 and above) are ignored for learning because they already have jobs.
- A CC can control one parameter; learning it again on another control moves the mapping.
- Mappings are saved with your DAW project (plug-in state), not inside presets.
- MIDI-learned controls do not include modulation amounts: use a macro for that.

16 Themes, Scaling and Settings

Make it fit your screen and your eyes

16.1 Light and dark themes

The default is the **dark indigo** theme with cyan, magenta and amber accents. Use the **Light / Dark** button in the top bar (it is labelled with the theme it will switch to) or **Menu > Settings > Dark theme / Light theme**. The choice is remembered.

Colour	Meaning on screen
Cyan	Oscillators, the modulation chips for Mod slots, knob values.
Magenta	Filters, macros, selection and drag highlights.
Amber	Performers, the Insert, active input ports and the learn ring.
Green	Noise, performance source chips and healthy meter level.

16.2 Interface size

Choose **Menu > Settings > Interface size** and pick 75, 100, 125, 150, 175 or 200 percent, or **drag the corner of the window**: the shape stays fixed and the size runs from 75 to 200 percent. Opening the keyboard makes the window taller but keeps the same scale. The size is remembered.

> **TIP.** A larger interface also makes it easier to draw Performer bars and to hit small knobs.

16.3 Where settings are stored

Theme, size, keyboard visibility and keyboard octaves are stored in **~/Library/Application Support/Phil S Audio/Massive PSXL.settings**. Everything about the sound itself is stored in presets and in the DAW project.

16.4 Undo and Redo

The Undo and Redo buttons (and the Menu entries) cover knob edits, routing changes, modulation assignments and removals, and macro renames. Loading a preset or Init clears the history.

16.5 Tooltips and messages

The oscillator **+** button has a tooltip. Most feedback is given as a short on-screen message (toast) below the top bar: assignments, imports, saves and refusals.

17 NKS Status

What is and is not implemented, stated plainly

NKS (Native Kontrol Standard) is a standard, owned by Native Instruments, that lets compatible hardware and software browse presets and control plug-ins through eight-knob pages. This chapter is the one place in the manual that names it, and it is deliberately factual.

> **IMPORTANT. Massive PSXL is not an officially NKS-certified product.** It does not carry the NKS logo, is not listed in any NKS browser, and ships no .nksf preset files. Nothing here has been tested with NKS hardware.

17.1 What is implemented in the plug-in

- **Parameter exposure.** Every sound parameter is a normal host-automatable parameter organised into groups (VST3 units) that follow the eight-parameter page idea: both Macro groups come first, then Oscillator 1, Oscillator 2, Noise, Filter 1, Filter 2, Insert + Amp, Voice + Master, the amp envelope, eight modulators, three Performers and the effects.
- **Smart macro naming.** A macro takes the name of the knob it controls, and the plug-in tells the host that parameter information changed, so hosts and controllers that read parameter names can show 'Filter 1 Cutoff' instead of 'Macro 1' (chapter 12). Names are saved in presets and plug-in state.
- **A page map.** The file **nks/psxl_nks_pages.json** in the project lists 24 intended eight-parameter controller pages (Macros 1-8 and 9-16 first, then Oscillators, Noise + Filter 1, Filter 2, Insert + Amp, Voice, Amp Envelope, Mod 2 to Mod 9, three Performers, EQ, Delay + Reverb and Chorus + Dist).
- **Provisional artwork.** Plug-in logos, banners and eight category tiles (bass, leads, pads, plucks, keys, drums, fx, sequence) are in Resources/art/nks. The sizes are placeholders and must be checked against the official requirements.
- **Categories.** Presets are organised as Bass, Leads, Pads, Plucks, Keys, Drums, FX and Sequence.

17.2 What is not possible without the standard's owner

- The **NKS developer kit, the NKS logo and a partner listing** are issued by Native Instruments on application. Until a product is approved it cannot use the NKS badge or appear in their browser.
- **.nksf preset files** (the preset format used by NKS browsers) and the exact artwork specifications come from that kit. Once available, the factory presets can be converted using the page map and artwork described above.

17.3 What to expect today

In NKS-capable host software the plug-in should behave as a normal VST3 or Audio Unit in generic plug-in mode, where parameter pages follow the host's parameter order. Because the macros are first, the first pages of controls are the macros with their automatic names. This is the intended behaviour; it has not been verified on hardware.

TIP. Whatever your hardware, you can get the benefit today: assign your eight favourite controls to Macros 1 to 8, let the automatic names appear, and map your controller's knobs to those macros (through your host's parameter mapping, or with MIDI Learn on the macro knobs).

18 Parameter Reference

Every parameter, range and default

Massive PSXL has **234 automatable parameters**. The tables list the parameter ID (as used by hosts and in preset files), the on-screen or host name, the range, the default and notes. The last column says whether the parameter can be a modulation target: **V** = voice modulation (any source), **G** = global (macros, mod wheel, aftertouch, pitch bend only), blank = not modulatable.

Discrete parameters show their choices in the Range column. The Mode of each filter is stored as a number (Mode 1 to Mode 4) and shown with names depending on the Model.

Macros (host group: Macros 1-8 and Macros 9-16 (two host groups))

ID	Name	Range	Default	Notes	Mod
macro1	Macro 1	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro2	Macro 2	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro3	Macro 3	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro4	Macro 4	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro5	Macro 5	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro6	Macro 6	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro7	Macro 7	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro8	Macro 8	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro9	Macro 9	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro10	Macro 10	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro11	Macro 11	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro12	Macro 12	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro13	Macro 13	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro14	Macro 14	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro15	Macro 15	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	
macro16	Macro 16	0 to 1	0	Display name changes at runtime (smart macro naming). Modulation SOURCE, not a target.	

Oscillator 1

ID	Name	Range	Default	Notes	Mod
o1_pos	Osc 1 Position	0 to 1	0	Scans through the frames of the selected wavetable.	V
o1_pitch	Osc 1 Pitch	-48 to +48 st	0	Whole-semitone steps on the knob; modulation is smooth (not quantised).	V
o1_fine	Osc 1 Fine	-100 to +100 ct	0		V
o1_level	Osc 1 Level	0 to 1	0.8	Level into the routing. Phase modulation still uses the oscillator at Level 0.	V
o1_uni	Osc 1 Unison	1 to 8	1	Number of stacked voices (integer).	
o1_detune	Osc 1 Detune	0 to 100 ct	15	Total unison spread (+/- cents across the stack). Mid-travel = 20 ct.	V
o1_pm	Osc 1 Phase Mod	0 to 1	0	Osc 2's output modulates Osc 1's phase (and Osc 1 modulates Osc 2).	V
o1_fold	Osc 1 Fold	0 to 1	0	Wavefolder after the oscillator.	V
o1_wt	Osc 1 Wavetable	0 to 31 (slot)	0	Slots 0-9 are built in (Morph Basic, Pulse Sweep, Harmonic Tilt, Vowel Choir, Hard Sync, FM Bells, Drawbar Organ, Glass Harmonics, Digital Grit, Fold Sweep); slots 10-31 hold imported tables.	

Oscillator 2

ID	Name	Range	Default	Notes	Mod
o2_pos	Osc 2 Position	0 to 1	0	Scans through the frames of the selected wavetable.	V
o2_pitch	Osc 2 Pitch	-48 to +48 st	0	Whole-semitone steps on the knob; modulation is smooth (not quantised).	V
o2_fine	Osc 2 Fine	-100 to +100 ct	0		V
o2_level	Osc 2 Level	0 to 1	0.8	Level into the routing. Phase modulation still uses the oscillator at Level 0.	V
o2_uni	Osc 2 Unison	1 to 8	1	Number of stacked voices (integer).	
o2_detune	Osc 2 Detune	0 to 100 ct	15	Total unison spread (+/- cents across the stack). Mid-travel = 20 ct.	V
o2_pm	Osc 2 Phase Mod	0 to 1	0	Osc 2's output modulates Osc 1's phase (and Osc 1 modulates Osc 2).	V
o2_fold	Osc 2 Fold	0 to 1	0	Wavefolder after the oscillator.	V
o2_wt	Osc 2 Wavetable	0 to 31 (slot)	1	Slots 0-9 are built in (Morph Basic, Pulse Sweep, Harmonic Tilt, Vowel Choir, Hard Sync, FM Bells, Drawbar Organ, Glass Harmonics, Digital Grit, Fold Sweep); slots 10-31 hold imported tables.	

Noise (host group: Noise + Filter 1)

ID	Name	Range	Default	Notes	Mod
n_level	Noise Level	0 to 1	0.5		V
n_color	Noise Color	0 to 1	0.5	Below 0.5 = low-pass, 0.5 = unfiltered, above 0.5 = high-pass.	V
n_type	Noise Type	White, Pink, Brown, Crackle	White		

Filter 1

ID	Name	Range	Default	Notes	Mod
f1_cut	Filter 1 Cutoff	20 to 20000 Hz	12000	Skewed: mid-travel = 1000 Hz. In Comb it sets the comb pitch, in Formant the vowel.	V
f1_res	Filter 1 Reso	0 to 1	0.15		V
f1_drive	Filter 1 Drive	0 to 1	0	Saturation ahead of the filter (about 1x to 8x gain).	V
f1_ktr	Filter 1 Key Track	-1 to +1	0	+1 = cutoff follows the keyboard 1:1 around note 60 (middle C).	
f1_model	Filter 1 Model	Ladder, State Variable, Comb, Formant	Ladder		
f1_mode	Filter 1 Mode	Mode 1 to Mode 4	Mode 1	Names depend on the Model: Ladder = LP 24, LP 12, BP, HP 24; State Variable = LP, BP, HP, Notch; Comb = Comb +, Comb -, Flange, Pluck; Formant = Male, Female, Child, Alien.	

Filter 2

ID	Name	Range	Default	Notes	Mod
f2_cut	Filter 2 Cutoff	20 to 20000 Hz	8000	Skewed: mid-travel = 1000 Hz. In Comb it sets the comb pitch, in Formant the vowel.	V
f2_res	Filter 2 Reso	0 to 1	0.15		V
f2_drive	Filter 2 Drive	0 to 1	0	Saturation ahead of the filter (about 1x to 8x gain).	V
f2_ktr	Filter 2 Key Track	-1 to +1	0	+1 = cutoff follows the keyboard 1:1 around note 60 (middle C).	
f2_model	Filter 2 Model	Ladder, State Variable, Comb, Formant	Ladder		
f2_mode	Filter 2 Mode	Mode 1 to Mode 4	Mode 1	Names depend on the Model: Ladder = LP 24, LP 12, BP, HP 24; State Variable = LP, BP, HP, Notch; Comb = Comb +, Comb -, Flange, Pluck; Formant = Male, Female, Child, Alien.	

Insert and Amp (host group: Insert + Amp)

ID	Name	Range	Default	Notes	Mod
ins_type	Insert Type	TanH, Fold, Bitcrush, Hard Clip, Rectify, Sine	TanH		
ins_drive	Insert Drive	0 to 1	0.3		V
ins_mix	Insert Mix	0 to 1	0.5	Dry/wet within the insert.	V
amp_level	Amp Level	0 to 1	0.7	Per-voice level after the amp envelope.	V
amp_pan	Amp Pan	-1 to +1	0	Equal-power pan.	V
amp_fb	Amp Feedback	0 to 1	0	Feeds a saturated copy of the voice output back into Filter 1 (or the Insert).	V

Voice and Master (host group: Voice + Master)

ID	Name	Range	Default	Notes	Mod
master_vol	Volume	0 to 1	0.7	Master gain follows volume squared (x2): 0.7 is about unity, 1.0 is +6 dB.	G
pitch	Pitch	-48 to +48 st	0	Global transpose in continuous semitones.	V
glide	Glide Time	0 to 5 s	0	Skewed: mid-travel = 0.4 s.	
glide_expo	Glide Expo	Off / On	Off	Exponential glide curve instead of constant time.	
voice_mode	Voice Mode	Poly, Mono, Legato	Poly		
voices	Polyphony	1 to 16	8	Maximum simultaneous voices in Poly mode.	
bend_range	Bend Range	0 to 24 st	2	Pitch wheel range.	
vel_sens	Velocity Sens	0 to 1	0.5	0 = every note at full level; 1 = fully velocity-controlled.	

Amp Envelope

ID	Name	Range	Default	Notes	Mod
m1_att	Amp Env Attack	0.5 ms to 10 s	5 ms	Skewed: mid-travel = 0.3 s. Linear ramp.	
m1_dec	Amp Env Decay	1 ms to 10 s	0.4 s	Skewed: mid-travel = 0.4 s. Exponential.	
m1_sus	Amp Env Sustain	0 to 1	0.8		
m1_rel	Amp Env Release	1 ms to 10 s	0.25 s	Skewed: mid-travel = 0.4 s. Exponential.	
m1_dly	Amp Env Delay	0 to 5 s	0	Wait before the attack starts. Skewed: mid-travel = 0.3 s.	
m1_hold	Amp Env Hold	0 to 5 s	0	Time held at the peak. Skewed: mid-travel = 0.3 s.	
m1_level	Amp Env Level	0 to 1	1	Scales the output of the slot (shown as Env Level for envelopes).	
m1_oneshot	Amp Env One Shot	Off / On	Off	Envelope ignores key release and runs through to its release stage.	

Modulator 2

The eight modulator slots share the same layout. Slot 2 is shown here as the example: for Mod 3 to Mod 9 replace the prefix m2_ with m3_ ... m9_ and 'Mod 2' with 'Mod 3' ... 'Mod 9'. All slots default to Type = Envelope. The Amp Envelope uses prefix m1_ and has no Type, Rate, Shape, Phase, Fade In, Bipolar, Sync or Division.

ID	Name	Range	Default	Notes	Mod
m2_type	Mod 2 Type	Envelope, LFO	Envelope	Chooses what the slot does.	
m2_att	Mod 2 Attack	0.5 ms to 10 s	10 ms	Skewed: mid-travel = 0.3 s. Linear ramp.	
m2_dec	Mod 2 Decay	1 ms to 10 s	0.4 s	Skewed: mid-travel = 0.4 s. Exponential.	
m2_sus	Mod 2 Sustain	0 to 1	0.5		
m2_rel	Mod 2 Release	1 ms to 10 s	0.3 s	Skewed: mid-travel = 0.4 s. Exponential.	
m2_dly	Mod 2 Delay	0 to 5 s	0	Wait before the attack starts. Skewed: mid-travel = 0.3 s.	
m2_hold	Mod 2 Hold	0 to 5 s	0	Time held at the peak. Skewed: mid-travel = 0.3 s.	
m2_level	Mod 2 Level	0 to 1	1	Scales the output of the slot (shown as Env Level for envelopes).	

ID	Name	Range	Default	Notes	Mod
m2_oneshot	Mod 2 One Shot	Off / On	Off	Envelope ignores key release and runs through to its release stage.	
m2_rate	Mod 2 Rate	0.01 to 50 Hz	2	LFO speed when Sync is off. Skewed: mid-travel = 2 Hz.	
m2_shape	Mod 2 Shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine		
m2_phase	Mod 2 Phase	0 to 1	0	Start phase at note-on.	
m2_fade	Mod 2 Fade In	0 to 10 s	0	Fades the LFO in after note-on. Skewed: mid-travel = 0.5 s.	
m2_bipolar	Mod 2 Bipolar	Off / On	On	On = -1 to +1; Off = 0 to +1.	
m2_sync	Mod 2 Sync	Off / On	Off	Locks the LFO to host tempo.	
m2_div	Mod 2 Division	1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T	1/4	Used when Sync is on.	

Performer 1

ID	Name	Range	Default	Notes	Mod
p1_steps	Perf 1 Steps	1 to 16	8	Loop length.	
p1_rate	Perf 1 Rate	0.1 to 40 Hz	4	Steps per second when Sync is off. Mid-travel = 4 Hz.	
p1_sync	Perf 1 Sync	Off / On	On	Locks the step rate to the host tempo.	
p1_div	Perf 1 Division	1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T	1/16	Length of one step when Sync is on.	
p1_smooth	Perf 1 Smooth	0 to 1	0	Glide between steps; 1 = glide across the whole step.	

Performer 2

ID	Name	Range	Default	Notes	Mod
p2_steps	Perf 2 Steps	1 to 16	8	Loop length.	
p2_rate	Perf 2 Rate	0.1 to 40 Hz	4	Steps per second when Sync is off. Mid-travel = 4 Hz.	
p2_sync	Perf 2 Sync	Off / On	On	Locks the step rate to the host tempo.	
p2_div	Perf 2 Division	1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T	1/16	Length of one step when Sync is on.	
p2_smooth	Perf 2 Smooth	0 to 1	0	Glide between steps; 1 = glide across the whole step.	

Performer 3

ID	Name	Range	Default	Notes	Mod
p3_steps	Perf 3 Steps	1 to 16	8	Loop length.	
p3_rate	Perf 3 Rate	0.1 to 40 Hz	4	Steps per second when Sync is off. Mid-travel = 4 Hz.	
p3_sync	Perf 3 Sync	Off / On	On	Locks the step rate to the host tempo.	
p3_div	Perf 3 Division	1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T	1/16	Length of one step when Sync is on.	
p3_smooth	Perf 3 Smooth	0 to 1	0	Glide between steps; 1 = glide across the whole step.	

EQ and Distortion (host group: EQ + Distortion)

ID	Name	Range	Default	Notes	Mod
eq_lo_g	EQ Low Gain	-18 to +18 dB	0	Low shelf.	G
eq_lo_f	EQ Low Freq	40 to 400 Hz	120		
eq_mid_g	EQ Mid Gain	-18 to +18 dB	0	Peaking band.	G
eq_mid_f	EQ Mid Freq	200 to 8000 Hz	1000	Skewed: mid-travel = 1500 Hz.	G
eq_mid_q	EQ Mid Q	0.3 to 6	1		
eq_hi_g	EQ High Gain	-18 to +18 dB	0	High shelf.	G
eq_hi_f	EQ High Freq	2000 to 16000 Hz	6000		
dist_drive	Dist Drive	0 to 1	0.3		G
dist_mix	Dist Mix	0 to 1	0	Distortion is bypassed at 0.	G

Delay, Reverb and Chorus (host group: Delay + Reverb + Chorus)

ID	Name	Range	Default	Notes	Mod
dly_time	Delay Time	1 to 2000 ms	350	Skewed: mid-travel = 300 ms. Ignored while Sync is on.	
dly_sync	Delay Sync	Off / On	Off		
dly_div	Delay Division	1/1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T	1/8	Used when Sync is on.	
dly_fb	Delay Feedback	0 to 0.95	0.35		G
dly_mix	Delay Mix	0 to 1	0		G
rev_size	Reverb Size	0 to 1	0.5		G
rev_damp	Reverb Damping	0 to 1	0.5		
rev_mix	Reverb Mix	0 to 1	0		G
cho_rate	Chorus Rate	0.05 to 8 Hz	0.6	Skewed: mid-travel = 0.8 Hz.	G
cho_depth	Chorus Depth	0 to 1	0.5		G
cho_mix	Chorus Mix	0 to 1	0		G

18.1 Quick counts

Item	Count
Parameters	234
Modulation sources	34 (9 envelope/LFO slots including the amp envelope, 3 Performers, 16 macros, Velocity, Key Track, Random, Mod Wheel, Aftertouch, Pitch Bend)
Modulation destinations	42 (28 voice, 14 global)
Maximum modulation assignments	256
Routing nodes / possible wires	7 nodes; 6 sources (Osc 1, Osc 2, Noise, Filter 1, Filter 2, Insert) and 4 destinations (Filter 1, Filter 2, Insert, Out); 14 possible wires
Wavetable slots	32 (10 built in, 22 user)
Wavetable frame size	2048 samples, up to 256 frames
Voices	up to 16

19 Mouse and Keyboard Reference

Every gesture in one place

Where	Gesture	Result
Any knob	Drag up / right (down / left)	Increase (decrease) the value
Any knob	Mouse wheel	Nudge the value
Any knob	Hover	Label turns into the exact value and unit
Any knob	Right-click	MIDI Learn, Clear MIDI mapping, Reset to default; Edit amount / Remove for each modulation source
Any knob	Drop a chip on it	Create a modulation assignment (amount +0.5)
Modulation chip	Drag onto a knob	Assign (a crosshair follows the pointer)
Macro chip (M1 to M16)	Double-click or right-click	Rename macro; Reset name
Wavetable display	Click / drag	Set Pos
Wavetable display	Drop an audio file	Import it as a wavetable on the open oscillator
Routing view	Drag from an output port to an input port	Create a wire
Routing view	Click a wire, then Delete or Backspace	Remove the wire
Routing view	Double-click a wire	Remove it
Routing view	Right-click a wire	Remove connection; Gain 25, 50, 75, 100, 150%
Routing view	Right-click empty space	Quick setups; Clear all connections
Envelope display	Drag the three handles	Attack; Decay and Sustain; Release
Performer bars	Click or drag	Set step heights
Preset list	Click; double-click; Return	Load; load and close; load
Top bar < >	Click	Previous / next preset
Preset name button	Click	Open / close the browser
On-screen keys	Click (height sets velocity)	Play a note
Pitch / Mod wheels	Drag	Pitch bend (springs back) / mod wheel (stays)

Where	Gesture	Result
Window corner	Drag	Resize (75 to 200 percent, fixed shape)
Dialogs	Return / Escape	OK or Save / Cancel

NOTE. Massive PSXL does not define its own global keyboard shortcuts. Your host may provide shortcuts of its own for Undo / Redo.

20 Troubleshooting

When something does not behave

Symptom	Likely cause	What to do
No sound at all after Load Init (Blank)	Init (Blank) has no routing.	Open the Routing tab and connect a source through to Out, or right-click empty space and choose a quick setup (chapter 5).
No sound, but wires exist	A node in the chain has no input, a Level or Amp Level is 0, Volume is 0, or the amp envelope is silent (Sustain 0 with a very short Decay).	Follow the glowing wires; check the node activity meters; raise Level, Amp Level and Volume.
Drums only blip, or cut off when I release the key	The amp envelope is not OneShot, or its Sustain is above 0.	Turn OneShot on in the Amp Env tab and set Sustain to 0.
A modulation does nothing	The source is voice-type and the target is a master effect, or the amount is 0, or the source is idle (an envelope with Sustain 0, or a Performer with Steps at zero height).	Check the toast. For effects use a macro, mod wheel or aftertouch. Right-click the knob and check Edit amount.
Toast: 'Voice modulators can't control global effects'	You dropped an envelope, LFO, Performer, Velocity, Key Track or Random on a master-effect knob.	Use a macro (and drive the macro from a controller or automation), mod wheel or aftertouch.
Toast: 'Signal flows left to right...'	You tried to connect backwards or to a source node.	Connect only to a later node: Osc/Noise > Filter 1 > Filter 2 > Insert > Out.
The plug-in does not appear in my DAW	Not installed, wrong folder, or the host has not rescanned.	Check the install paths in chapter 2 and rescan plug-ins. Logic runs its own Audio Unit validation for new components.
macOS says the plug-in or installer cannot be opened	Ad-hoc signed, not notarised, and quarantined by a download.	Control-click the installer and choose Open, use Privacy and Security > Open Anyway, or clear the quarantine flag as shown in chapter 2.
Crackling, dropouts or high CPU	Too many unison voices, high polyphony or a small audio buffer.	Reduce Unison, Polyphony or the number of held notes; raise the buffer size.
Stuck note	A missing note-off or sustain pedal state.	Send All Notes Off from the host or stop and restart playback; release the sustain pedal.
Delay or LFO timing is wrong	No host tempo is available.	Without a host tempo (for example in the standalone app) synced features use 120 BPM. Use a DAW tempo or turn Sync off.
Wavetable import fails ('Could not read that audio file')	Unsupported or damaged file.	Use WAV, AIFF, FLAC or OGG. Make sure the file is longer than a few dozen samples.
Imported wavetable sounds like noise	The file is not a single-cycle waveform; it is being sliced into 2048-sample frames.	Prepare a single cycle, or use Pos to find the musical region and treat it as a texture.

Symptom	Likely cause	What to do
A preset sounds different from my last session	A wavetable that the preset needs was not saved, or tuning was changed.	Factory presets use only built-in tables. Check the tuning in the Voice tab and reset to 12-TET if needed.
Macro name does not update on my controller	The host does not refresh parameter names live.	Reload the plug-in window or re-select the plug-in in the host's parameter list.
Pitch arpeggio is slightly out of tune	Performer steps are continuous values drawn by hand.	Increase the interface size, or write exact values into a .psxlperf pattern file (chapter 11, tutorial 22).
Interface too small or too large	Display scaling.	Menu > Settings > Interface size, or drag the window corner.

21 Glossary

Terms used in this manual

Term	Meaning
Amp envelope	The envelope (slot 1) that shapes each note's loudness.
Amount	How strongly a source moves a destination, -1 to +1, as a fraction of the knob's travel.
Assignment	A source-to-destination modulation link with an amount.
Band-limited	Containing no harmonics above half the sample rate, which avoids aliasing.
Bipolar	Swinging both above and below zero (-1 to +1).
Chip	The small crosshair control you drag onto a knob to assign a modulation source.
Comb filter	A short feedback delay that makes regularly spaced resonances.
Cutoff	The frequency where a filter begins to act (the vowel selector in the Formant model).
Destination	A knob that can be modulated.
Detune	The spread in cents between unison voices.
Division	A musical note length (1/4, 1/8T and so on) used by synced LFOs, Performers and the delay.
Envelope	A shape that evolves over a note: delay, attack, hold, decay, sustain, release.
Fold	A wavefolder that reflects a wave back on itself.
Formant	A resonant peak of the vocal tract; three of them make a vowel.
Frame	One single-cycle waveform inside a wavetable.
Glide	Portamento: sliding pitch between notes.
Init (Blank)	The default sound with no routing and no modulation.
Insert	The per-voice waveshaping node.
Key Track	Making a filter or modulation follow the pitch of the note played.
Legato	A voice mode where overlapping notes change pitch without retriggering.
LFO	Low-frequency oscillator, a repeating modulation shape.
Macro	A knob that you assign to other knobs; it takes the name of the first target.
Mip level	A progressively band-limited version of a wavetable used at higher pitches.
Node	A box in the Routing view (Osc, Noise, Filter, Insert, Out).
One Shot	An envelope that ignores key release.
Performer	A 16-step modulation sequencer.

Term	Meaning
Phase Mod	Phase modulation of one oscillator by the other (FM-like).
Port	The circle on a node's edge where wires start and end.
Reso	Resonance: emphasis at the filter's cutoff.
Routing	Which sources feed which processors, shown as wires.
Sample and Hold	An LFO shape that picks a random value each cycle and holds it.
Scala	A file format (.scl) describing musical scales for microtuning.
Source	Anything that modulates: envelopes, LFOs, Performers, macros, controllers.
Unison	Several detuned copies of an oscillator played together.
Wavetable	A set of frames scanned by the Pos control.
Wire	A connection between two node ports. It glows with the passing audio.

PART II

Tutorials

Twenty-eight recipes that start from Load Init (Blank): drums, bass, leads, keys, pads, effects, sequenced sounds, macros and wavetable import.

How to Use the Tutorials

Recipes that start from Load Init (Blank)

Each tutorial builds one sound from scratch, so you learn the controls by using them. They all start the same way:

1. **Menu > Load Init (Blank)**: every parameter returns to its default, and there is **no routing**.
2. **Routing** tab: connect the nodes the recipe lists. Either right-click empty space and choose a quick setup, or drag from an output port (right edge) to an input port (left edge).
3. Set the oscillator, noise, filter, insert and envelope values from the steps and the settings table.
4. Assign modulation by dragging a chip onto a knob, then right-click the knob > **Edit amount** and type or drag to the amount given.
5. Add the effects, play, and adjust by ear.

Reading the recipe tables

Column	Meaning
Where	The panel or tab: Osc 1, Osc 2, Noise, Filter 1, Insert, Amp, Amp Env, Mod 2 and so on. 'Mod 2 (LFO)' means the Mod 2 tab with Type set to LFO.
Parameter	The labels on screen, in the order listed. A slash separates several parameters in one row.
Value	The setting in the same order. 'Pos 0.66' means turn Pos to 0.66; you can hover over a knob while dragging to see the exact value.

STARTING POINT. Every number in these tutorials is a **starting point**, not a rule. Real patches are found by ear: change Cutoff, Decay and amounts until it sounds right for your track.

TIP. To compare with a ready-made version, load the factory preset named at the top of each tutorial. Use **Menu > Load Init (Blank)** afterwards to start again.

NOTE. Quick tab reminders: oscillator knobs belong to whichever of **Osc 1 | Osc 2** is selected, the filter knobs to **Filter 1 | Filter 2**, and the effects to the **EQ / Delay / Verb / Chor / Dist** tabs. Make the destination knob visible before dropping a chip on it.

The tutorials

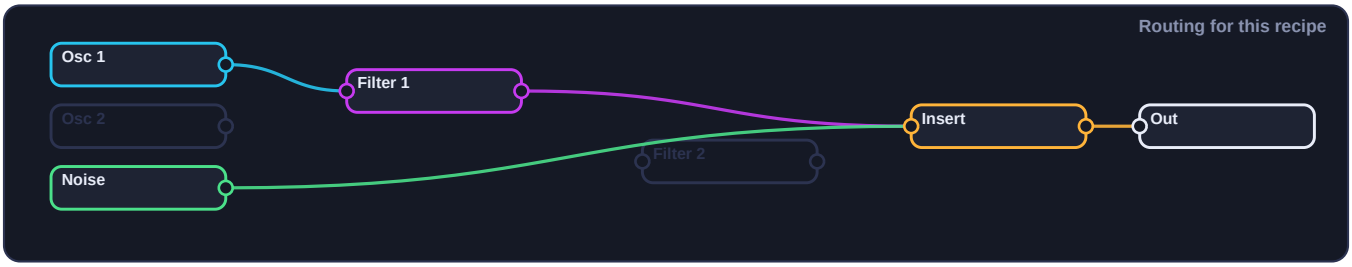
Group	Tutorials
Drums	1. Kick drum, 2. Snare drum, 3. Closed and open hi-hats, 4. Hand clap, 5. Tom
Bass	6. Sub bass, 7. Wobble bass, 8. Acid bass, 9. Reese bass
Leads	10. Supersaw lead, 11. Mono lead (expressive)
Plucks and Keys	12. Pluck, 13. Bell, 14. Electric piano, 15. Drawbar organ
Pads	16. String pad, 17. Choir and vowel pad, 18. Brass stab
Effects	19. Riser and sweep, 20. Laser zap, 21. Wind and ocean
Sequenced and special	22. Arpeggio with a Performer, 23. Trance gate with a Performer, 24. Talking filter (formant), 25. FM bell with Phase Mod, 26. Tempo-synced stutter
Workflow	27. Build your own macro-controlled patch, 28. Import your own wavetable

Tutorial 1: Kick drum

Drums | Factory preset to compare: Kick Tight, Kick Punch (punchier), Kick Deep 808 (long and low)

A punchy sine kick: a sine tone whose pitch falls quickly, plus a touch of noise for the click.

Routing: Osc 1 > Filter 1 > Insert > Out, and Noise > Insert (the click skips the filter). Draw the wires by hand: drag from the right-edge port of each node to the left-edge port of the next.



Steps

1. **Menu > Load Init (Blank)**. Open the **Routing** tab and draw the four wires listed above.
2. Select **Osc 1**. Set the wavetable to **Morph Basic** and **Pos** to 0 (pure sine). Set **Level** 0.95.
3. Open the **Amp Env** tab: Attack 0.001, Decay 0.28, Sustain 0, Release 0.08 and switch **OneShot** on, so the drum rings out even on a short tap.
4. Open the **Mod 2** tab (Type: Envelope): Attack 0.0005, Decay 0.11, Sustain 0, Release 0.05, OneShot on. Drag the **Mod 2** chip onto the Osc 1 **Pitch** knob, then right-click the knob > **Edit amount** and set +0.28.
5. Set the Noise, Filter 1 and Insert values from the table. Play low notes (C0 to C1 on the keyboard).

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Morph Basic / 0
Osc 1	Pitch / Level	0 / 0.95
Noise	Level / Color / Type	0.10 / 0.85 / White
Filter 1	Model / Mode / Cutoff / Reso	Ladder / LP 24 / 9000 Hz / 0.05
Insert	Type / Drive / Mix	TanH / 0.25 / 1.0
Amp Env	Attack / Decay / Sustain / Release	0.001 / 0.28 / 0 / 0.08 (OneShot on)
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.0005 / 0.11 / 0 / 0.05 (OneShot on)

Modulation assignments

Source	Target	Amount	Why
Mod 2	Osc 1 Pitch	+0.28	About 27 semitones of downward sweep in the first 100 ms: the 'thump'.

Playing tips and variations

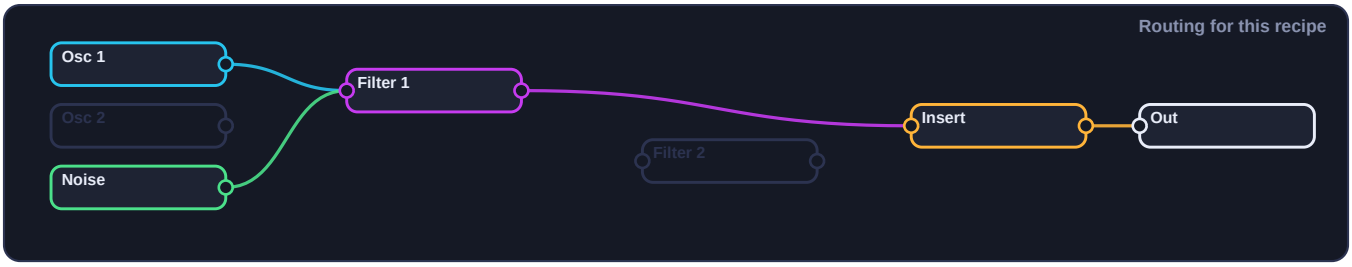
- For a long 808-style kick set Osc 1 **Pitch** to -12, Amp Env Decay to 0.9 and Insert Drive to 0.15 (that is the factory *Kick Deep 808*).
- More click: raise Noise Level to 0.2 or add Filter 1 Cutoff modulation. Less click: lower Noise Level to 0.
- The pitch sweep depth sets the character: 0.2 is soft and round, 0.33 is hard and snappy.

Tutorial 2: Snare drum

Drums | Factory preset to compare: Snare Crack, Snare Tight

A band-passed noise burst with a short sine body underneath.

Routing: Noise and Osc 1 > Filter 1 > Insert > Out (quick setup 'Noise > Filter 1 > Insert > Out', then add Osc 1 > Filter 1).



Steps

1. **Menu > Load Init (Blank)**. Routing tab: right-click an empty spot in the Routing tab > **Noise > Filter 1 > Insert > Out (drums / FX)**. Then drag a wire from **Osc 1** to **Filter 1**.
2. Noise panel: Level 0.9, Type White, Color 0.55. Osc 1: **Morph Basic**, Pos 0, Pitch +12, Level 0.5.
3. Filter 1: choose Model **State Variable**, Mode **BP**, Cutoff 2400, Reso 0.35.
4. Insert: Type **Fold**, Drive 0.2, Mix 1.0.
5. Amp Env: Attack 0.0008, Decay 0.22, Sustain 0, Release 0.05, OneShot on.

Settings

Where	Parameter	Value (starting point)
Noise	Level / Color / Type	0.9 / 0.55 / White
Osc 1	Wavetable / Pos / Pitch / Level	Morph Basic / 0 / +12 / 0.5
Filter 1	Model / Mode / Cutoff / Reso	State Variable / BP / 2400 Hz / 0.35
Insert	Type / Drive / Mix	Fold / 0.2 / 1.0
Amp Env	Attack / Decay / Sustain / Release	0.0008 / 0.22 / 0 / 0.05 (OneShot on)

Playing tips and variations

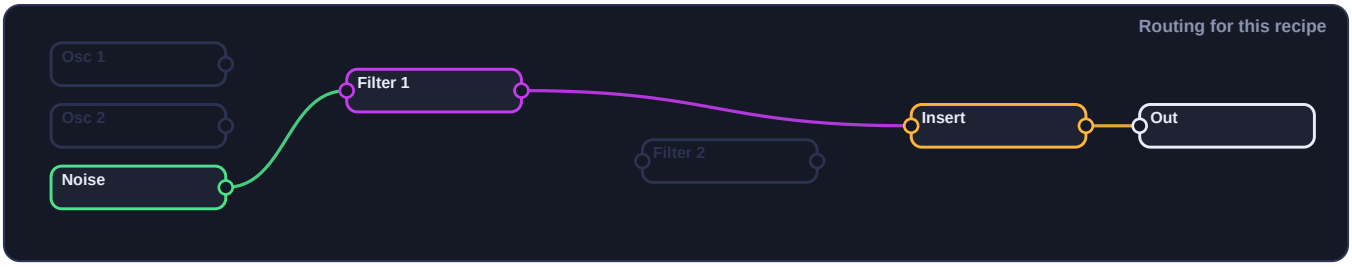
- Shorter and tighter (*Snare Tight*): Decay 0.14, Cutoff 3200, Osc 1 Pitch +16, Osc 1 Level 0.4.
- A little reverb (Verb tab, Mix 0.15) makes it sit in a room; a delay on a snare is rarely wanted.
- A Rim Shot is the same recipe with Decay 0.05, Cutoff 1800, Reso 0.55, Osc 1 Pitch +24, Level 0.6 (factory *Rim Shot*).

Tutorial 3: Closed and open hi-hats

Drums | Factory preset to compare: Hat Closed, Hat Open, Shaker

High-passed noise with a very short (closed) or longer (open) envelope.

Routing: Noise > Filter 1 > Insert > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing tab: right-click an empty spot in the Routing tab > **Noise > Filter 1 > Insert > Out (drums / FX)**.
2. Noise: Level 0.9, Type White, Color 0.9 (the Color knob above 0.5 is a high-pass).
3. Filter 1: **State Variable** / **HP**, Cutoff 9000, Reso 0.15.
4. Insert: leave Mix at 0 for a clean hat (or Type Fold, Drive 0.3, Mix 0.3 for a metallic edge).
5. **Closed hat**: Amp Env Attack 0.0008, Decay 0.045, Sustain 0, Release 0.05, OneShot on. **Open hat**: Decay 0.35, Release 0.2 (save this variation as a second preset).

Settings

Where	Parameter	Value (starting point)
Noise	Level / Color / Type	0.9 / 0.9 / White
Filter 1	Model / Mode / Cutoff / Reso	State Variable / HP / 9000 Hz / 0.15 (open hat: 8000 Hz)
Insert	Mix	0 (transparent)
Amp Env (closed)	Attack / Decay / Sustain / Release	0.0008 / 0.045 / 0 / 0.05, OneShot on
Amp Env (open)	Attack / Decay / Sustain / Release	0.0008 / 0.35 / 0 / 0.2, OneShot on

Playing tips and variations

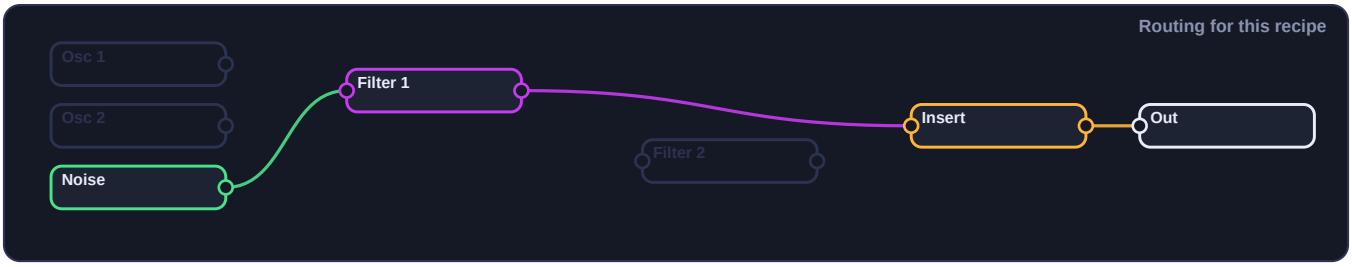
- There is no hat choke group: an open hat rings for its full Decay even if you then play a closed hat. Keep Decay and Release short enough for your tempo, or set Voice Mode to Mono on the open-hat patch so repeated hits cut the previous one.
- Shaker: Decay 0.09, Hold 0.02, Cutoff 7000, Reso 0.3, Color 0.8 (factory *Shaker*).
- Metallic hat: set Noise Type to **Crackle** or Filter 1 Model to **Comb** with Cutoff 700 and Reso 0.8.

Tutorial 4: Hand clap

Drums | Factory preset to compare: Clap Room

A noise burst whose level is flickered by a Performer so it sounds like several hands.

Routing: Noise > Filter 1 > Insert > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Noise > Filter 1 > Insert > Out (drums / FX)**.
2. Noise: Level 0.9, Color 0.6, Type White. Filter 1: **State Variable** / **BP**, Cutoff 1500, Reso 0.45. Insert Mix 0.
3. Amp panel: set **Level** to 0 (the performer will supply the level). Amp Env: Attack 0.0008, Decay 0.16, Sustain 0, Release 0.05, OneShot on.
4. Open **Perf 1**: Steps 6, Sync off, Rate 28 Hz, Smooth 0.3. Draw the bars roughly: 1.0, 0.05, 0.85, 0.05, 0.7, 0.35 (full height, nearly empty, 85 percent, nearly empty, 70 percent, a third).
5. Drag the **Perf 1** chip onto the Amp **Level** knob and set the amount to +0.80. Play a note: the clap now stutters 3 times then tails off.

Settings

Where	Parameter	Value (starting point)
Noise	Level / Color / Type	0.9 / 0.6 / White
Filter 1	Model / Mode / Cutoff / Reso	State Variable / BP / 1500 Hz / 0.45
Amp	Level	0 (performer takes over)
Amp Env	Attack / Decay / Sustain / Release	0.0008 / 0.16 / 0 / 0.05, OneShot on
Perf 1	Steps / Sync / Rate / Smooth	6 / off / 28 Hz / 0.3

Modulation assignments

Source	Target	Amount	Why
Perf 1	Amp Level	+0.80	Each step sets the loudness of one hand-clap flam.

Playing tips and variations

- The factory *Clap Room* uses the synced alternative: Sync on, Division 1/32, 8 steps. Sync off with a Rate near 30 Hz gives tighter flams.
- Add Reverb Mix 0.2 for a roomy clap. Add a second Performer on Filter 1 Cutoff for variation between hands.

Tutorial 5: Tom

Drums | Factory preset to compare: Tom Low, Tom High

Like the kick but higher, with a gentler sweep and a longer tone.

Routing: Osc 1 > Filter 1 > Out (quick setup 'Osc 1 > Filter 1 > Out').



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0, Pitch +5, Level 0.95. Filter 1: Cutoff 9000, Reso 0.05.
3. Amp Env: Attack 0.001, Decay 0.5, Sustain 0, Release 0.08, OneShot on.
4. Mod 2 (Envelope): Attack 0.0005, Decay 0.145, Sustain 0, Release 0.05, OneShot on. Drag the Mod 2 chip onto Osc 1 **Pitch**, amount +0.12.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Pitch / Level	Morph Basic / 0 / +5 / 0.95
Filter 1	Cutoff / Reso	9000 Hz / 0.05
Amp Env	Attack / Decay / Sustain / Release	0.001 / 0.5 / 0 / 0.08, OneShot on
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.0005 / 0.145 / 0 / 0.05, OneShot on

Modulation assignments

Source	Target	Amount	Why
Mod 2	Osc 1 Pitch	+0.12	About 11 semitones of sweep.

Playing tips and variations

- High tom (*Tom High*): Osc 1 Pitch +17, Decay 0.35. Play a fill across several keys for a tuned set.
- Add Noise Level 0.05 (Color 0.8) routed to Filter 1 for a stick attack.

Tutorial 6: Sub bass

Bass | Factory preset to compare: Sub Pillar, Deep Weight

A clean low sine with a second sine an octave below.

Routing: Osc 1 + Osc 2 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 + Osc 2 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0. Osc 2: change the wavetable to **Morph Basic** (it defaults to Pulse Sweep), Pos 0, Pitch -12, Level 0.5.
3. Filter 1: Ladder / LP 24, Cutoff 380, Reso 0.1.
4. Amp Env: Attack 0.004, Decay 0.3, Sustain 0.95, Release 0.18.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Level	Morph Basic / 0 / 0.8
Osc 2	Wavetable / Pos / Pitch / Level	Morph Basic / 0 / -12 / 0.5
Filter 1	Model / Mode / Cutoff / Reso	Ladder / LP 24 / 380 Hz / 0.1
Amp Env	Attack / Decay / Sustain / Release	0.004 / 0.3 / 0.95 / 0.18

Playing tips and variations

- *Deep Weight* adds harmonics so small speakers can hear it: Osc 1 Pos 0.12, Osc 2 Level 0.7, Cutoff 300, plus Insert (TanH, Drive 0.35, Mix 0.6) after Filter 1.
- Set Voice Mode to **Mono** (Pitch panel) with Glide 0.05 for sliding 808 style lines.
- Check the bass on small speakers: sines below 60 Hz are inaudible on laptops without harmonic content.

Tutorial 7: Wobble bass

Bass | Factory preset to compare: Wobble Machine, Dub Wobble

A saw bass whose filter is swept by a tempo-synced LFO.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66 (saw), Unison 3, Detune 10.
3. Filter 1: Ladder / LP 24, Cutoff 520, Reso 0.5, Drive 0.2. Amp Env: Attack 0.003, Decay 0.3, Sustain 0.9, Release 0.15.
4. Open **Mod 2**, set Type to **LFO**, Shape **Sine**, switch **Sync** on and choose Division **1/8**.
5. Drag the Mod 2 chip onto Filter 1 **Cutoff** and set the amount to +0.34. Play low notes.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 3 / 10 ct
Filter 1	Model / Mode / Cutoff / Reso / Drive	Ladder / LP 24 / 520 Hz / 0.5 / 0.2
Amp Env	Attack / Decay / Sustain / Release	0.003 / 0.3 / 0.9 / 0.15
Mod 2 (LFO)	Shape / Sync / Division / Bipolar	Sine / on / 1/8 / on

Modulation assignments

Source	Target	Amount	Why
Mod 2 (LFO)	Filter 1 Cutoff	+0.34	The cutoff swings both ways around the knob position, in time with the host tempo.

Playing tips and variations

- Dub version (*Dub Wobble*): Shape Triangle, Division 1/4, Reso 0.55, Cutoff 450, and Delay Mix 0.25 (Sync on, Division 1/8).
- Faster or slower wobble: change the Division (1/16 is aggressive, 1/2 is slow). With Sync off use Rate in Hz.
- If the wobble sounds too thin lower the base Cutoff; to hear more of the bottom raise the amount instead.
- In a DAW the synced LFO follows the project tempo. When no host tempo is available (for example in the standalone app) tempo-synced features assume 120 BPM.

Tutorial 8: Acid bass

Bass | Factory preset to compare: Acid Line, Mono Acid Lead

Resonant ladder filter with a fast envelope and legato slides.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66 (saw). Pitch panel: Voice Mode **Legato**, Glide 0.07.
3. Filter 1: Ladder / LP 24, Cutoff 560, Reso 0.72, Drive 0.35. Amp Env: Attack 0.002, Decay 0.25, Sustain 0.85, Release 0.1.
4. Mod 2 (Envelope): Attack 0.002, Decay 0.2, Sustain 0, Release 0.15. Drag its chip onto Filter 1 **Cutoff**, amount +0.34.
5. Play a pattern with overlapping notes: legato notes slide and do not retrigger the envelopes; detached notes re-pluck the filter.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Morph Basic / 0.66
Pitch panel	Voice Mode / Glide	Legato / 0.07 s
Filter 1	Model / Mode / Cutoff / Reso / Drive	Ladder / LP 24 / 560 Hz / 0.72 / 0.35
Amp Env	Attack / Decay / Sustain / Release	0.002 / 0.25 / 0.85 / 0.1
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.002 / 0.2 / 0 / 0.15

Modulation assignments

Source	Target	Amount	Why
Mod 2	Filter 1 Cutoff	+0.34	Each retriggered note snaps the filter open then closes it.

Playing tips and variations

- Accents: drag the **Velocity** chip (left of the macro strip) onto Filter 1 Cutoff at +0.2 and play some notes harder.
- Higher Reso gives the squelch; near the top of the range the ladder rings almost like a sine.
- Add Delay (Sync on, Division 1/8, Mix 0.2) and Dist Mix 0.2 for the classic dirty acid sound.

Tutorial 9: Reese bass

Bass | Factory preset to compare: Reese Beast

Two heavily detuned saw stacks beating against each other, with slow filter movement.

Routing: Osc 1 + Osc 2 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 + Osc 2 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66, Unison 4, Detune 24. Osc 2: **Morph Basic**, Pos 0.66, Fine +9, Unison 4, Detune 20.
3. Filter 1: Ladder / LP 24, Cutoff 1000, Reso 0.2. Amp Env: Attack 0.01, Decay 0.4, Sustain 0.9, Release 0.3.
4. Mod 2: Type **LFO**, Shape **Triangle**, Rate 0.3 Hz (Sync off). Drag its chip onto Filter 1 Cutoff, amount +0.12.
5. Chorus tab: Mix 0.25 to widen. Play low, long notes.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 4 / 24 ct
Osc 2	Wavetable / Pos / Fine / Unison / Detune	Morph Basic / 0.66 / +9 ct / 4 / 20 ct
Filter 1	Model / Mode / Cutoff / Reso	Ladder / LP 24 / 1000 Hz / 0.2
Amp Env	Attack / Decay / Sustain / Release	0.01 / 0.4 / 0.9 / 0.3
Mod 2 (LFO)	Shape / Rate	Triangle / 0.3 Hz
FX: Chorus	Mix	0.25

Modulation assignments

Source	Target	Amount	Why
Mod 2 (LFO)	Filter 1 Cutoff	+0.12	Slow breathing of the brightness.

Playing tips and variations

- The movement comes from the beating between the two detuned stacks; change Osc 2 Fine to alter the beat speed.
- Add Insert (Fold or TanH, Mix 0.4) after Filter 1 for the growl (connect Filter 1 > Insert > Out).

Tutorial 10: Supersaw lead

Leads | Factory preset to compare: Super Stack, Detuned Trance

A big, wide stack of eight detuned saws.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66, **Unison** 8, **Detune** 34.
3. Filter 1: Ladder / LP 24, Cutoff 7000, Reso 0.15. Amp Env: Attack 0.006, Decay 0.3, Sustain 0.85, Release 0.35.
4. FX: Chorus Mix 0.25; Reverb Size 0.6, Mix 0.2.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 8 / 34 ct
Filter 1	Model / Mode / Cutoff / Reso	Ladder / LP 24 / 7000 Hz / 0.15
Amp Env	Attack / Decay / Sustain / Release	0.006 / 0.3 / 0.85 / 0.35
FX	Chorus Mix / Reverb Size / Reverb Mix	0.25 / 0.6 / 0.2

Playing tips and variations

- Trance version (*Detuned Trance*): Osc 1 Unison 7, Detune 28; add Osc 2 (Morph Basic, Pos 0.66) at Pitch +12, Unison 3, Detune 14, Level 0.35; Delay Sync on 1/8, Mix 0.3.
- Each unison voice has a random starting phase, so no two notes sound exactly alike.
- Playing chords with 8 unison voices uses 8 oscillators per note: lower Polyphony if your computer struggles.

Tutorial 11: Mono lead (expressive)

Leads | Factory preset to compare: Classic Saw Lead, Expressive Lead

A single-voice lead with portamento, plus mod wheel and aftertouch expression.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66, Unison 3, Detune 14. Pitch panel: Voice Mode **Mono**, Glide 0.03.
3. Filter 1: Ladder / LP 24, Cutoff 5000, Reso 0.18. Amp Env: Attack 0.004, Decay 0.3, Sustain 0.85, Release 0.3.
4. FX: Delay Sync on, Division 1/8, Feedback 0.35, Mix 0.2; Reverb Mix 0.15.
5. Expression: drag the **ModWhl** chip (macro strip, left) onto Osc 1 **Pos** (+0.4) and onto Filter 1 **Cutoff** (+0.25); drag **AftTch** onto Cutoff (+0.15).
6. Vibrato: Mod 3 Type LFO, Shape Sine, Rate 5.2 Hz, Fade In 0.6. Drag its chip onto the Pitch panel **Pitch** knob with amount +0.003 (about a third of a semitone).

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 3 / 14 ct
Pitch panel	Voice Mode / Glide	Mono / 0.03 s
Filter 1	Model / Mode / Cutoff / Reso	Ladder / LP 24 / 5000 Hz / 0.18 (Expressive: 2500 Hz / 0.25)
Amp Env	Attack / Decay / Sustain / Release	0.004 / 0.3 / 0.85 / 0.3
Mod 3 (LFO)	Shape / Rate / Fade In	Sine / 5.2 Hz / 0.6 s
FX	Delay (Sync, 1/8, Fb 0.35) / Reverb	Mix 0.2 / Mix 0.15

Modulation assignments

Source	Target	Amount	Why
Mod Wheel	Osc 1 Position	+0.40	Wheel morphs the timbre.
Mod Wheel	Filter 1 Cutoff	+0.25	Wheel opens the filter.
Aftertouch	Filter 1 Cutoff	+0.15	Pressing harder brightens.
Mod 3 (LFO)	Pitch	+0.003	Delayed vibrato (the Fade In ramps it up).

Playing tips and variations

- Mono mode retriggers every note; choose **Legato** to slide between overlapping notes without re-striking the envelopes.
- The on-screen keyboard has mod and pitch wheels. Aftertouch needs hardware that sends channel pressure.
- Pitch bend range is set in the Voice tab (Bend Range, default 2 semitones).

Tutorial 12: Pluck

Plucks and Keys | Factory preset to compare: Classic Pluck, Soft Pluck, Detuned Pluck

A short saw note whose filter snaps shut: the basis of plucks, stabs and arps.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66. Filter 1: Cutoff 700, Reso 0.2.
3. Amp Env: Attack 0.001, Decay 0.45, Sustain 0, Release 0.25.
4. Mod 2 (Envelope): Attack 0.001, Decay 0.25, Sustain 0, Release 0.2. Drag its chip onto Filter 1 **Cutoff**, amount +0.40.
5. FX: Delay Sync on, Division 1/8, Mix 0.15; Reverb Mix 0.2.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Morph Basic / 0.66
Filter 1	Cutoff / Reso	700 Hz / 0.2
Amp Env	Attack / Decay / Sustain / Release	0.001 / 0.45 / 0 / 0.25
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.001 / 0.25 / 0 / 0.2
FX	Delay (Sync, 1/8) / Reverb	Mix 0.15 / Mix 0.2

Modulation assignments

Source	Target	Amount	Why
Mod 2	Filter 1 Cutoff	+0.40	Bright attack that darkens quickly.

Playing tips and variations

- Detuned Pluck: Unison 4, Detune 16, Cutoff 900. Soft Pluck: Pos 0.3, Cutoff 600, amount +0.30.
- Because Sustain is 0 the note decays even when you hold the key; Decay is the pluck length.
- EDM Stab: Unison 6, Detune 26, Decay 0.28, Cutoff 2000 (factory *EDM Stab*).

Tutorial 13: Bell

Plucks and Keys | Factory preset to compare: Bell Pluck, Glass Pluck

A bright inharmonic strike from the FM Bells wavetable, ringing out.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: wavetable **FM Bells**, Pos 0.4. Filter 1: Cutoff 6000, Reso 0.12.
3. Amp Env: Attack 0.001, Decay 0.9, Sustain 0, Release 0.5.
4. Mod 2 (Envelope): Attack 0.001, Decay 0.4, Sustain 0, Release 0.3, chip onto Filter 1 **Cutoff**, amount +0.10.
5. FX: Reverb Mix 0.2; Delay Sync on, Division 1/8, Mix 0.15.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	FM Bells / 0.4
Filter 1	Cutoff / Reso	6000 Hz / 0.12
Amp Env	Attack / Decay / Sustain / Release	0.001 / 0.9 / 0 / 0.5
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.001 / 0.4 / 0 / 0.3

Modulation assignments

Source	Target	Amount	Why
Mod 2	Filter 1 Cutoff	+0.10	Softens the strike as the note rings.

Playing tips and variations

- Move Osc 1 **Pos** from 0 (soft, near-sine) to 1 (bright, harsh) to change the bell's character.
- Glass Pluck: wavetable Glass Harmonics, Pos 0.35, Cutoff 7000, Decay 0.8.
- Play high notes: bells sit best two to three octaves above middle C.

Tutorial 14: Electric piano

Plucks and Keys | Factory preset to compare: Warm E-Piano, Soft Rhodes, Tine Keys

Soft FM tines with velocity-controlled brightness and a gentle chorus.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: wavetable **FM Bells**, Pos 0.12. Filter 1: Cutoff 3500, Reso 0.1. Open the **Voice** tab and set **Vel Sens** to 0.9.
3. Amp Env: Attack 0.002, Decay 1.2, Sustain 0, Release 0.5.
4. Mod 2 (Envelope): Attack 0.001, Decay 0.5, Sustain 0.2, Release 0.3; chip onto Filter 1 **Cutoff** at +0.15.
5. Drag the **Velocity** chip onto Osc 1 **Pos** at +0.2: harder keys give brighter tines.
6. FX: Chorus Mix 0.3, Reverb Mix 0.2.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	FM Bells / 0.12
Filter 1	Cutoff / Reso	3500 Hz / 0.1
Voice tab	Vel Sens	0.9
Amp Env	Attack / Decay / Sustain / Release	0.002 / 1.2 / 0 / 0.5
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.001 / 0.5 / 0.2 / 0.3
FX	Chorus Mix / Reverb Mix	0.3 / 0.2

Modulation assignments

Source	Target	Amount	Why
Mod 2	Filter 1 Cutoff	+0.15	Tine attack.
Velocity	Osc 1 Position	+0.20	Touch-sensitive brightness.

Playing tips and variations

- Sustain 0 gives a piano-like decay even while holding the key; lengthen Decay for a more sustained electric piano.
- Soft Rhodes: Pos 0.08, Cutoff 2800, Decay 1.5, Velocity onto Filter 1 Cutoff +0.2 instead of Position.

Tutorial 15: Drawbar organ

Plucks and Keys | Factory preset to compare: Drawbar Organ, Rock Organ

The Drawbar Organ wavetable turns the position knob into a registration control.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: wavetable **Drawbar Organ**, Pos 0.3. Filter 1: Cutoff 9000, Reso 0.
3. Amp Env: Attack 0.003, Decay 0.2, Sustain 1.0, Release 0.08 (organs have no decay).
4. FX: Chorus Mix 0.3 (stand-in for the rotary speaker), Reverb Mix 0.15, Dist Drive 0.3, Dist Mix 0.15.
5. Drag macro chip **M1** onto Osc 1 **Pos**. M1 is renamed 'Osc 1 Position': turn it live to change registrations.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Drawbar Organ / 0.3
Filter 1	Cutoff / Reso	9000 Hz / 0
Amp Env	Attack / Decay / Sustain / Release	0.003 / 0.2 / 1.0 / 0.08
FX	Chorus Mix / Reverb Mix / Dist Drive / Dist Mix	0.3 / 0.15 / 0.3 / 0.15

Modulation assignments

Source	Target	Amount	Why
Macro 1	Osc 1 Position	+0.5 (default)	Live registration control; the macro takes the knob's name.

Playing tips and variations

- Rock Organ: Pos 0.7, Dist Drive 0.5, Dist Mix 0.4, Chorus Mix 0.2.
- Position morphs through different mixes of the 1st, 2nd, 3rd, 4th, 6th, 8th, 10th, 12th and 16th harmonics.
- Add key click: Noise Level 0.05, Color 0.9, routed to Filter 1, with a very short Mod envelope on Noise Level.

Tutorial 16: String pad

Pads | Factory preset to compare: Warm Strings, Wide Saw Wash, Velvet

A slow-attack saw ensemble with chorus and a long reverb.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66, Unison 6, Detune 22. Filter 1: Cutoff 2400, Reso 0.1.
3. Amp Env: Attack 0.8, Decay 0.6, Sustain 0.85, Release 1.6.
4. FX: Chorus Mix 0.35; Reverb Size 0.75, Mix 0.3.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 6 / 22 ct
Filter 1	Cutoff / Reso	2400 Hz / 0.1
Amp Env	Attack / Decay / Sustain / Release	0.8 / 0.6 / 0.85 / 1.6
FX	Chorus Mix / Reverb Size / Reverb Mix	0.35 / 0.75 / 0.3

Playing tips and variations

- Add Osc 2 (Morph Basic, Pos 0.66) a fifth up or an octave down at a lower Level for body.
- A slow filter envelope (Mod 2: Attack 1.5, amount +0.35 on Cutoff) makes the pad 'open' as you hold it (factory *Sunrise*).
- Drag M1 onto Filter 1 Cutoff and M2 onto Reverb Mix to play the pad's brightness and space.

Tutorial 17: Choir and vowel pad

Pads | Factory preset to compare: Choir Dream, Evolving Vowels, Aurora

The Vowel Choir wavetable scanned slowly by an LFO.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: wavetable **Vowel Choir**, Pos 0.3, Unison 5, Detune 14. Filter 1: Cutoff 7000, Reso 0.15.
3. Amp Env: Attack 0.8, Decay 0.6, Sustain 0.85, Release 1.6.
4. Mod 2: Type **LFO**, Shape **Sine**, Rate 0.2 Hz. Chip onto Osc 1 **Pos**, amount +0.12.
5. FX: Reverb Size 0.85, Mix 0.45.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Vowel Choir / 0.3 / 5 / 14 ct
Filter 1	Cutoff / Reso	7000 Hz / 0.15
Amp Env	Attack / Decay / Sustain / Release	0.8 / 0.6 / 0.85 / 1.6
Mod 2 (LFO)	Shape / Rate	Sine / 0.2 Hz
FX: Reverb	Size / Mix	0.85 / 0.45

Modulation assignments

Source	Target	Amount	Why
Mod 2 (LFO)	Osc 1 Position	+0.12	Drifts between neighbouring vowels.

Playing tips and variations

- Position 0, 0.25, 0.5, 0.75 and 1.0 are the vowels A, E, I, O, U. The vowel shapes are baked into the waveform, so they shift with the note you play: they sound most natural in the low-to-middle register.
- Evolving Vowels: Pos 0, LFO Rate 0.07 Hz, amount +0.8 sweeps the whole vowel range slowly.

Tutorial 18: Brass stab

Pads | Factory preset to compare: Cinematic Brass Lead (slower), EDM Stab (shorter)

Two saws with a filter envelope that opens quickly: brass swells and stabs.

Routing: Osc 1 + Osc 2 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 + Osc 2 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66, Unison 4, Detune 12. Osc 2: **Morph Basic**, Pos 0.66, Fine +6, Unison 2, Level 0.5.
3. Filter 1: Ladder / LP 24, Cutoff 1200, Reso 0.15.
4. Amp Env: Attack 0.03, Decay 0.25, Sustain 0.7, Release 0.15.
5. Mod 2 (Envelope): Attack 0.04, Decay 0.3, Sustain 0.35, Release 0.2; chip onto Filter 1 **Cutoff**, amount +0.35.
6. FX: Reverb Mix 0.2.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 4 / 12 ct
Osc 2	Wavetable / Pos / Fine / Unison / Level	Morph Basic / 0.66 / +6 ct / 2 / 0.5
Filter 1	Cutoff / Reso	1200 Hz / 0.15
Amp Env	Attack / Decay / Sustain / Release	0.03 / 0.25 / 0.7 / 0.15
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.04 / 0.3 / 0.35 / 0.2

Modulation assignments

Source	Target	Amount	Why
Mod 2	Filter 1 Cutoff	+0.35	The 'blat' as the brass opens up.

Playing tips and variations

- For a slow orchestral swell (*Cinematic Brass Lead* style) use Amp Attack 0.1, Mod 2 Attack 0.12, Decay 0.4, Sustain 0.6 and a bigger reverb.
- Drag Velocity onto Filter 1 Cutoff (+0.2) so harder hits sound more brassy.

Tutorial 19: Riser and sweep

Effects | Factory preset to compare: Riser, Downlifter

A long build: a noisy saw stack whose filter and pitch rise over several seconds.

Routing: Osc 1 and Noise > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**, then drag a wire from **Noise** to **Filter 1**.
2. Osc 1: wavetable **Digital Grit**, Pos 0.2, Unison 4, Detune 40. Noise: Level 0.35, Color 0.8, Type White.
3. Filter 1: Ladder / LP 24, Cutoff 300, Reso 0.4. Amp Env: Attack 1.0, Decay 0.5, Sustain 1.0, Release 1.0.
4. Mod 2 (Envelope): Attack 6.0, Decay 0.1, Sustain 1.0, Release 0.5. Chip onto Filter 1 **Cutoff** (+0.70) and Osc 1 **Pitch** (+0.15).
5. FX: Reverb Mix 0.35. Hold a note for at least 6 seconds.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Digital Grit / 0.2 / 4 / 40 ct
Noise	Level / Color / Type	0.35 / 0.8 / White
Filter 1	Cutoff / Reso	300 Hz / 0.4
Amp Env	Attack / Decay / Sustain / Release	1.0 / 0.5 / 1.0 / 1.0
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	6.0 / 0.1 / 1.0 / 0.5

Modulation assignments

Source	Target	Amount	Why
Mod 2	Filter 1 Cutoff	+0.70	The sweep upwards.
Mod 2	Osc 1 Pitch	+0.15	About 14 semitones of rise.

Playing tips and variations

- Downlifter: Cutoff 9000, Mod 2 Attack 0.001, Decay 4.0, Sustain 0, amounts -0.6 (Cutoff) and -0.12 (Pitch), Amp Env Decay 4.0.
- To make the riser fit the music: attack seconds = beats x 60 / BPM (8 beats at 120 BPM = 4 s). The longest attack is 10 s.

Tutorial 20: Laser zap

Effects | Factory preset to compare: Laser Zap

A very fast downward pitch sweep on a hard-sync tone.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Osc 1: wavetable **Hard Sync**, Pos 0.4. Filter 1: Cutoff 9000, Reso 0.3.
3. Amp Env: Attack 0.001, Decay 0.3, Sustain 0, Release 0.1.
4. Mod 2 (Envelope): Attack 0.0005, Decay 0.15, Sustain 0, Release 0.1. Chip onto Osc 1 **Pitch**, amount +0.35.
5. FX: Delay Sync on, Division 1/8, Mix 0.25.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Hard Sync / 0.4
Filter 1	Cutoff / Reso	9000 Hz / 0.3
Amp Env	Attack / Decay / Sustain / Release	0.001 / 0.3 / 0 / 0.1
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.0005 / 0.15 / 0 / 0.1

Modulation assignments

Source	Target	Amount	Why
Mod 2	Osc 1 Pitch	+0.35	About 34 semitones of fall in 150 ms.

Playing tips and variations

- Reverse the sign (-0.35) for an upward 'power-up' blip.
- Longer Mod 2 Decay (0.4) makes slower, more cartoonish zaps; higher Osc 1 Pos makes a more metallic zap.

Tutorial 21: Wind and ocean

Effects | Factory preset to compare: Wind Texture, Ocean Wash

Filtered noise with a slow random filter movement and a long envelope.

Routing: Noise > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: drag one wire from the Noise out-port to the Filter 1 in-port, and one from Filter 1 to Out.
2. **Wind**: Noise Level 0.9, Type **Pink**, Color 0.4. Filter 1: Cutoff 900, Reso 0.4. Amp Env: Attack 1.5, Decay 0.5, Sustain 1.0, Release 2.0.
3. Mod 2: Type **LFO**, Shape **Smooth Random**, Rate 0.13 Hz. Chip onto Filter 1 **Cutoff**, amount +0.30.
4. FX: Reverb Size 0.9, Mix 0.4.
5. **Ocean**: Type **Brown**, Color 0.35, Cutoff 600, Reso 0.2, Amp Attack 2.0 / Release 3.0, LFO Shape Sine at 0.09 Hz, amount +0.35, Reverb Mix 0.45.

Settings

Where	Parameter	Value (starting point)
Noise (wind)	Level / Type / Color	0.9 / Pink / 0.4
Filter 1 (wind)	Cutoff / Reso	900 Hz / 0.4
Amp Env	Attack / Decay / Sustain / Release	1.5 / 0.5 / 1.0 / 2.0
Mod 2 (LFO)	Shape / Rate	Smooth Random / 0.13 Hz
Noise (ocean)	Type / Color	Brown / 0.35

Modulation assignments

Source	Target	Amount	Why
Mod 2 (LFO)	Filter 1 Cutoff	+0.30 (wind), +0.35 (ocean)	Slow gusting / surf swell.

Playing tips and variations

- Oscillators stay silent because they are not routed. You do not need to lower their levels.
- A stormier wind: raise Filter 1 Reso to 0.6 and the LFO amount to +0.45.

Tutorial 22: Arpeggio with a Performer

Sequenced and special | Factory preset to compare: Minor Arp, Major Arp

A step sequence that moves Osc 1's pitch every 1/16 note while you hold a single key.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**. Osc 1: **Morph Basic**, Pos 0.66. Filter 1: Cutoff 2600, Reso 0.25.
2. Amp Env: Attack 0.001, Decay 0.2, Sustain 0.8, Release 0.1. Mod 2 (Envelope): Attack 0.001, Decay 0.15, Sustain 0.2, Release 0.1, chip onto Filter 1 Cutoff at +0.25.
3. Open **Perf 1**. Steps 8, Sync on, Division **1/16**, Smooth 0.
4. Drag the **Perf 1** chip onto Osc 1 **Pitch** and edit the amount to +0.25. With this amount a full-height bar is +24 semitones.
5. Draw the bars (minor arpeggio): step heights 0, 0.125, 0.29, 0.5, 0.625, 0.5, 0.29, 0.125. That is 0, +3, +7, +12, +15, +12, +7, +3 semitones. Hold a key: the arpeggio climbs and falls.
6. FX: Delay Sync on, 1/8, Mix 0.25; Reverb Mix 0.2.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Morph Basic / 0.66
Filter 1	Cutoff / Reso	2600 Hz / 0.25
Amp Env	Attack / Decay / Sustain / Release	0.001 / 0.2 / 0.8 / 0.1
Perf 1	Steps / Sync / Division / Smooth	8 / on / 1/16 / 0
Perf 1 bars	Heights (steps 1 to 8)	0, 0.125, 0.29, 0.5, 0.625, 0.5, 0.29, 0.125

Modulation assignments

Source	Target	Amount	Why
Perf 1	Osc 1 Pitch	+0.25	Full bar = 24 semitones; height x 24 = semitones.
Mod 2	Filter 1 Cutoff	+0.25	A little pluck at the start of the note.

Playing tips and variations

- Performer pitch steps are continuous, not snapped to semitones, so drawn bars can be a few cents out. For exact values use **Patterns > Save pattern...** to create a .psxlperf file, edit it in a text editor so it contains *0,0.125,0.292,0.5,0.625,0.5,0.292,0.125,0,0,0,0,0,0,0,0*, then **Patterns > Load pattern...**. Setting the interface to 150 to 200 percent (Menu > Settings > Interface size) also helps when drawing.
- Quick shapes live in **Patterns**: Ramp up, Ramp down, Triangle, Alternate high / low, Random, Flat, All zero, Gate on/off pairs.
- The factory *Minor Arp* uses amount +0.125 (12 st per bar); its 15-semitone step exceeds the bar range and plays as 12 semitones.
- Each note has its own performer that restarts on note-on, so chords stay in step when played together.

Tutorial 23: Trance gate with a Performer

Sequenced and special | Factory preset to compare: Trance Gate Lead, Gate Pad

A rhythmic gate: a Performer opens and closes the amp level in 16th notes.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**. Osc 1: **Morph Basic**, Pos 0.66, Unison 6, Detune 24. Filter 1: Cutoff 6000, Reso 0.15.
2. Amp Env: Attack 0.005, Decay 0.3, Sustain 1.0, Release 0.2. In the **Amp** panel set **Level** to 0 (the performer will open it).
3. Open **Perf 1**: Steps 16, Sync on, Division 1/16, Smooth 0.15.
4. Draw the pattern (1 = full bar, 0 = empty): 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 0. Quickest route: **Patterns > Gate: on/off pairs** then adjust bars.
5. Drag the **Perf 1** chip onto the Amp **Level** knob; edit the amount to +0.75.
6. FX: Delay Sync on, 1/8, Mix 0.3; Reverb Mix 0.25. Hold a chord.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 6 / 24 ct
Filter 1	Cutoff / Reso	6000 Hz / 0.15
Amp	Level	0
Amp Env	Attack / Decay / Sustain / Release	0.005 / 0.3 / 1.0 / 0.2
Perf 1	Steps / Sync / Division / Smooth	16 / on / 1/16 / 0.15

Modulation assignments

Source	Target	Amount	Why
Perf 1	Amp Level	+0.75	Bars become the gate pattern; a small Smooth avoids clicks.

Playing tips and variations

- The gate pattern restarts on each note-on, so play notes on the beat to keep it in time.
- For a gentler pulse keep the empty bars at 0.2 instead of 0.
- *Gate Pad* uses 8 steps (1 1 0 1 1 0 1 0) and a pad sound instead of a lead.

Tutorial 24: Talking filter (formant)

Sequenced and special | Factory preset to compare: Talking Bass, Robot Voice

The Formant filter model sweeps through vowels as the Cutoff moves.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**. Osc 1: **Morph Basic**, Pos 0.66 (saw).
2. Filter 1: Model **Formant**, Mode **Male**, Cutoff 700, Reso 0.55. Amp Env: Attack 0.003, Decay 0.3, Sustain 0.9, Release 0.15.
3. Mod 2: Type **LFO**, Shape **Triangle**, Sync on, Division 1/4. Chip onto Filter 1 **Cutoff**, amount +0.22.
4. Play sustained notes: the sound says a repeating 'ah - eh - ee - oh'.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Morph Basic / 0.66
Filter 1	Model / Mode / Cutoff / Reso	Formant / Male / 700 Hz / 0.55
Amp Env	Attack / Decay / Sustain / Release	0.003 / 0.3 / 0.9 / 0.15
Mod 2 (LFO)	Shape / Sync / Division	Triangle / on / 1/4

Modulation assignments

Source	Target	Amount	Why
Mod 2 (LFO)	Filter 1 Cutoff	+0.22	Sweeps roughly from 'ah' toward 'oh' and back.

Playing tips and variations

- Cutoff maps to vowels logarithmically: lowest = A, about 110 Hz = E, about 630 Hz = I, about 3.5 kHz = O, highest = U.
- Modes: Male, Female, Child (all formants higher) and Alien (lower, with a shifted second formant).
- Robot voice (*Robot Voice*): wavetable Vowel Choir, Filter 1 Model Comb, Cutoff 300, Reso 0.7, and a Sample & Hold LFO on Osc 1 Pos.

Tutorial 25: FM bell with Phase Mod

Sequenced and special | Factory preset to compare: No factory preset uses Phase Mod. Compare Bell Pluck and Cowbell FM (FM Bells wavetable).

Osc 2 silently phase-modulates Osc 1: a classic two-operator FM bell whose brightness decays.

Routing: Osc 1 > Filter 1 > Out. Osc 2 is not routed: it works as a modulator only.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out** (only Osc 1 is heard).
2. Osc 1: wavetable **Morph Basic**, Pos 0 (sine), Level 0.8, **PhaseMod** 0.10. Phase Mod on Osc 1 means *Osc 2 modulates Osc 1*.
3. Osc 2: wavetable **Morph Basic**, Pos 0 (sine), Pitch +22. Level can stay at its default: Osc 2 is not routed, so it is inaudible, but it still modulates Osc 1.
4. Filter 1: Cutoff 9000, Reso 0.05. Amp Env: Attack 0.001, Decay 1.8, Sustain 0, Release 1.0.
5. Mod 2 (Envelope): Attack 0.001, Decay 0.6, Sustain 0, Release 0.4. Chip onto Osc 1 **PhaseMod**, amount +0.50: the FM index starts high and decays.
6. FX: Reverb Size 0.7, Mix 0.3.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / PhaseMod	Morph Basic / 0 / 0.10
Osc 2	Wavetable / Pos / Pitch	Morph Basic / 0 / +22 (about ratio 3.4:1)
Filter 1	Cutoff / Reso	9000 Hz / 0.05
Amp Env	Attack / Decay / Sustain / Release	0.001 / 1.8 / 0 / 1.0
Mod 2 (Envelope)	Attack / Decay / Sustain / Release	0.001 / 0.6 / 0 / 0.4
FX: Reverb	Size / Mix	0.7 / 0.3

Modulation assignments

Source	Target	Amount	Why
Mod 2	Osc 1 PhaseMod	+0.50	Bright inharmonic strike that mellows as the index decays.

Playing tips and variations

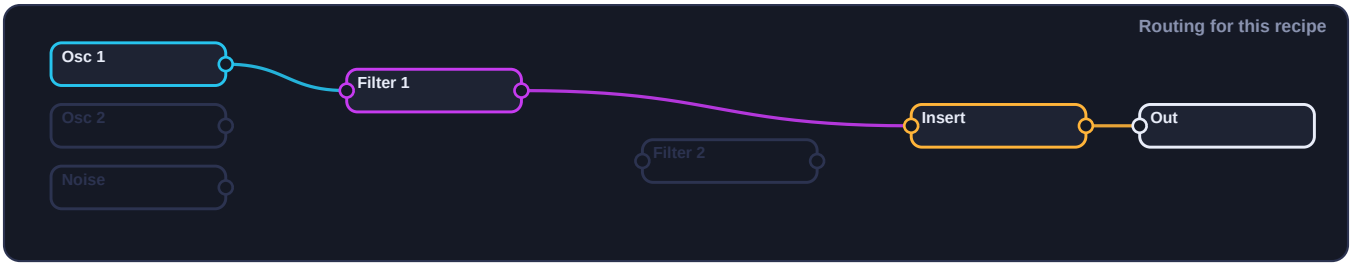
- Pitch +12 / +19 / +24 on Osc 2 gives harmonic ratios 2, 3, 4 (organ or e-piano). Odd values such as +22 or Osc 2 Fine give bells.
- Osc 2's own Phase Mod is driven by Osc 1 (feedback-style FM); leave it at 0 for a clean two-operator bell.
- Raise Osc 1 Fold for extra metallic edge.

Tutorial 26: Tempo-synced stutter

Sequenced and special | Factory preset to compare: Glitch Stutter

A square LFO chops the Amp Level into 1/16 note gates, with a bit-crusher for grit.

Routing: Osc 1 > Filter 1 > Insert > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing tab: draw Osc 1 > Filter 1, Filter 1 > Insert, Insert > Out.
2. Osc 1: wavetable **Digital Grit**, Pos 0.5. Filter 1: Cutoff 3000, Reso 0.3. Insert: Type **Bitcrush**, Drive 0.6, Mix 0.9.
3. Amp Env: Attack 0.003, Decay 0.3, Sustain 0.9, Release 0.2.
4. Mod 2: Type **LFO**, Shape **Square**, Sync on, Division 1/16, **Bipolar off**. Chip onto Amp **Level**, amount -0.90.
5. Hold a note: the sound chops on and off in 16th notes.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	Digital Grit / 0.5
Filter 1	Cutoff / Reso	3000 Hz / 0.3
Insert	Type / Drive / Mix	Bitcrush / 0.6 / 0.9
Amp Env	Attack / Decay / Sustain / Release	0.003 / 0.3 / 0.9 / 0.2
Mod 2 (LFO)	Shape / Sync / Division / Bipolar	Square / on / 1/16 / off

Modulation assignments

Source	Target	Amount	Why
Mod 2 (LFO)	Amp Level	-0.90	Unipolar square: the LFO pushes the level to zero for half of each cycle.

Playing tips and variations

- Change Division to 1/8 or 1/8T for different grooves; Phase shifts where in the bar the gate begins.
- Use Smooth Random or Sample & Hold on a second LFO onto Filter 1 Cutoff to vary each chop.

Tutorial 27: Build your own macro-controlled patch

Workflow | Factory preset to compare: Sweep Machine (macros set up for a bass)

Make a simple lead and put its best controls on macros, using automatic macro naming.

Routing: Osc 1 + Osc 2 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 + Osc 2 > Filter 1 > Out**.
2. Osc 1: **Morph Basic**, Pos 0.66, Unison 3, Detune 14. Osc 2: **Morph Basic**, Pos 0.66, Pitch -12, Level 0.5. Filter 1: Cutoff 1500, Reso 0.2. Amp Env: Attack 0.01, Decay 0.4, Sustain 0.8, Release 0.4.
3. Select the **Filter 1** tab so its Cutoff and Reso knobs are visible. Drag the **M1** chip (the little crosshair under the first macro knob) onto the **Cutoff** knob. Look at the toast: Macro 1 now controls Filter 1 Cutoff, and the macro knob is relabelled 'Filter 1 Cutoff'.
4. Drag **M2** onto **Reso**. Select the **Osc 1** tab and drag **M3** onto **Pos**. Open the **Verb** tab (right panel) and drag **M4** onto **Mix**: this is a global effect, which macros are allowed to control.
5. Right-click the Cutoff knob > **Edit amount: Macro 1** and set it to +0.50. Right-click Reso and set +0.25.
6. Drag **M5** onto Dist **Drive** and **M5** again onto Dist **Mix** (Dist tab). The macro keeps the name of the first target. Double-click the M5 chip (or right-click > Rename macro...) and call it 'Grit'.
7. Menu > Save preset... switches to the Browse tab; click **Save preset...**, give it a name, pick a category and Save. The macro names are saved with the preset.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos / Unison / Detune	Morph Basic / 0.66 / 3 / 14 ct
Osc 2	Wavetable / Pos / Pitch / Level	Morph Basic / 0.66 / -12 / 0.5
Filter 1	Cutoff / Reso	1500 Hz / 0.2
Amp Env	Attack / Decay / Sustain / Release	0.01 / 0.4 / 0.8 / 0.4

Modulation assignments

Source	Target	Amount	Why
Macro 1	Filter 1 Cutoff	+0.50	Macro auto-named 'Filter 1 Cutoff'.
Macro 2	Filter 1 Reso	+0.25	Auto-named 'Filter 1 Reso'.
Macro 3	Osc 1 Position	+0.5 (default)	Auto-named 'Osc 1 Position'.
Macro 4	Reverb Mix	+0.5 (default)	A global target.
Macro 5	Dist Drive and Dist Mix	+0.5 (default)	Renamed by you to 'Grit'.

Playing tips and variations

- Right-click a macro knob > **MIDI Learn**, then move a hardware knob: the macro follows it (mapped CCs are stored with your DAW project).

- A macro can drive many parameters at once with different amounts, including negative ones: this is how one knob can open the filter while closing the reverb.
- Remove an assignment from the target knob: right-click > Remove Macro 1. If you did not rename the macro by hand its name returns to 'Macro 1' when its last assignment is removed.

Tutorial 28: Import your own wavetable

Workflow | Factory preset to compare: Not applicable (user content)

Turn any audio file into a playable wavetable and morph through it with Pos.

Routing: Osc 1 > Filter 1 > Out.



Steps

1. **Menu > Load Init (Blank)**. Routing: right-click an empty spot in the Routing tab > **Osc 1 > Filter 1 > Out**.
2. Prepare a WAV, AIFF, FLAC or OGG file. Best results: a single cycle of 2048 samples (a one-frame table), or several 2048-sample cycles joined end to end (a multi-frame table).
3. Drop the file on the oscillator display (the waveform window in the Oscillator panel), or press the **+** button next to the wavetable menu, or use **Menu > Settings > Import wavetable...**. The toast says 'Imported ... into Osc N'.
4. The new table appears in the wavetable drop-down as 'User: filename' and is selected on the active oscillator. Sweep **Pos** through it.
5. Set Filter 1 and Amp Env to taste (try Cutoff 8000, Attack 0.005, Decay 0.4, Sustain 0.8, Release 0.3) and save the preset: the imported table is saved inside user presets and your DAW project.

Settings

Where	Parameter	Value (starting point)
Osc 1	Wavetable / Pos	User: your file / 0 to 1
Filter 1	Cutoff / Reso	8000 Hz / 0.15
Amp Env	Attack / Decay / Sustain / Release	0.005 / 0.4 / 0.8 / 0.3

Playing tips and variations

- How files are sliced: the file is mixed to mono and divided into frames of about 2048 samples (length divided by 2048, rounded, between 1 and 256 frames), each resampled to 2048 points. A short file becomes a single frame; a one-second 44.1 kHz recording becomes about 22 frames. The maximum is 256 frames, and very long files are read only up to about a million samples (roughly 24 seconds at 44.1 kHz) and squeezed into those 256 frames.
- Each frame is peak-normalised and its DC offset removed. Expect a texture, not a recording: the file is played as a looping cycle at the pitch of the key.
- There are 22 user slots (the drop-down shows them as 'User: ...'); when all are full the last one is replaced. The Voice tab lists user tables.

A Factory Presets

All 117 presets by category

The browser shows presets sorted by category, then name. Total: **117**.

Bass (20)

Acid Line	Dub Wobble	Reese Beast
Analog Growl	Electro Pluck Bass	Rubber Band
Bit Bass	FM Growler	Sub Pillar
Dark Rumble	Fold Sub	Sweep Machine
Deep Weight	Grit Bass	Talking Bass
Detuned Square Bass	Mono Lead Bass	Wobble Machine
Distorted Saw	Pulse Punch	

Leads (19)

Bright Pluck Lead	Fold Lead	Super Stack
Cinematic Brass Lead	Glass Lead	Sync Screamer
Classic Saw Lead	Hoover Lead	Theremin
Detuned Trance	Mono Acid Lead	Vowel Lead
Digital Edge	PWM Dreamer	Whistle
Expressive Lead	Retro Poly Lead	
FM Lead	Square Lead	

Pads (20)

Analog Poly Pad	Dust Drift	Shimmer Wash
Aurora	Evolving Vowels	Soft Piano Pad
Breathing Pad	Frozen Bells Pad	Sunrise
Cathedral Organ Pad	Glass Pad	Velvet
Choir Dream	Hollow Reed	Warm Strings
Cold Fusion	Motion Pad	Wide Saw Wash
Dark Ambience	Orbit	

Plucks (15)

Bell Pluck	Harp-ish	Pulse Pluck
Classic Pluck	Kalimba	Soft Pluck
Detuned Pluck	Koto	Steel Drum
EDM Stab	Marimba	Synth Strum
Glass Pluck	Pizzicato	Wood Block

Keys (10)

Clav Funk	Rock Organ	Warm E-Piano
Digital Piano	Soft Rhodes	Wurly Bite
Drawbar Organ	Tine Keys	
Glass Keys	Vox Keys	

Drums (15)

Clap Room	Kick Hard	Shaker
Cowbell FM	Kick Punch	Snare Crack
Hat Closed	Kick Sub Boom	Snare Tight
Hat Open	Kick Tight	Tom High
Kick Deep 808	Rim Shot	Tom Low

FX (10)

Alien Transmission	Metal Hit	Sci-Fi Drone
Downlifter	Ocean Wash	Wind Texture
Glitch Stutter	Riser	
Laser Zap	Robot Voice	

Sequence (8)

Bouncing Ball	Major Arp	Random Bleeps
Filter Stepper	Minor Arp	Trance Gate Lead
Gate Pad	Pulse Motion	